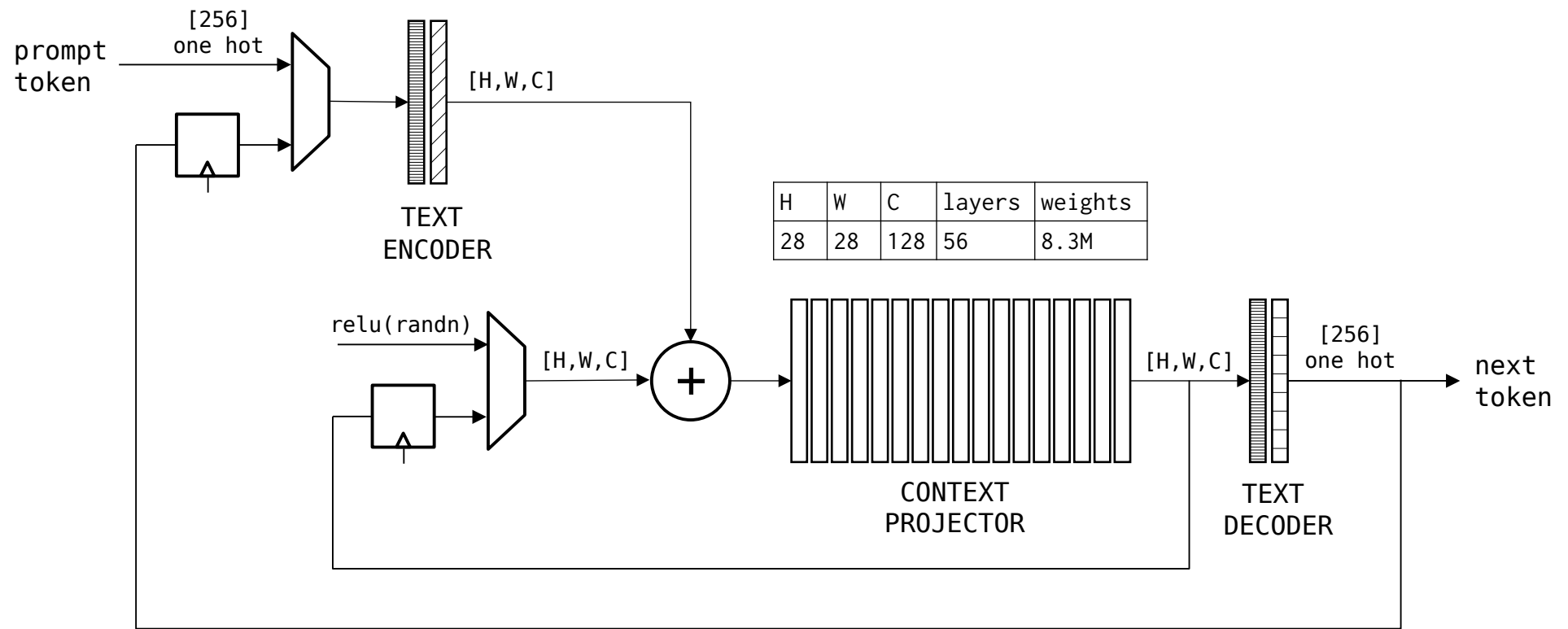
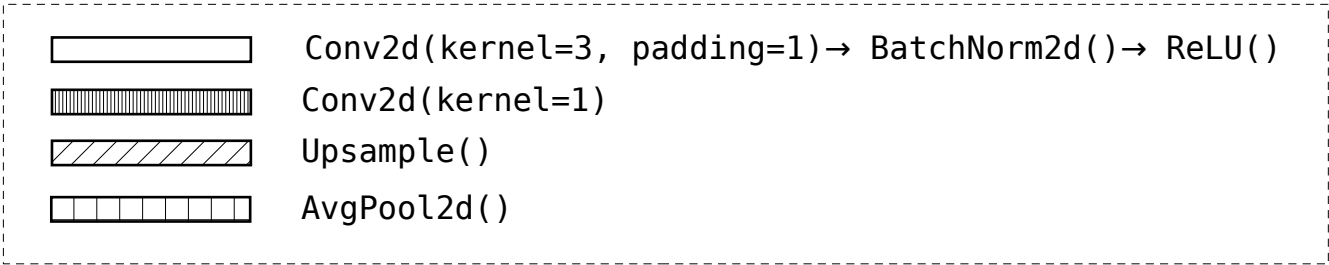
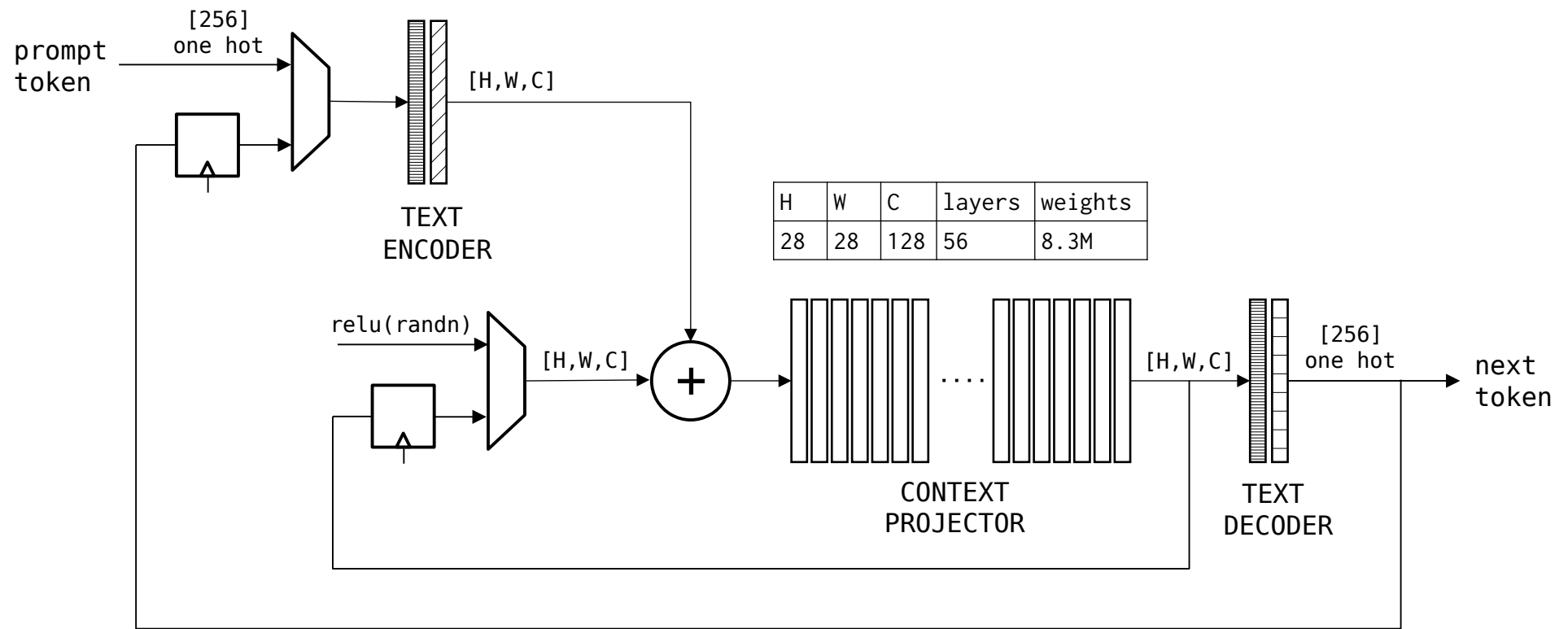
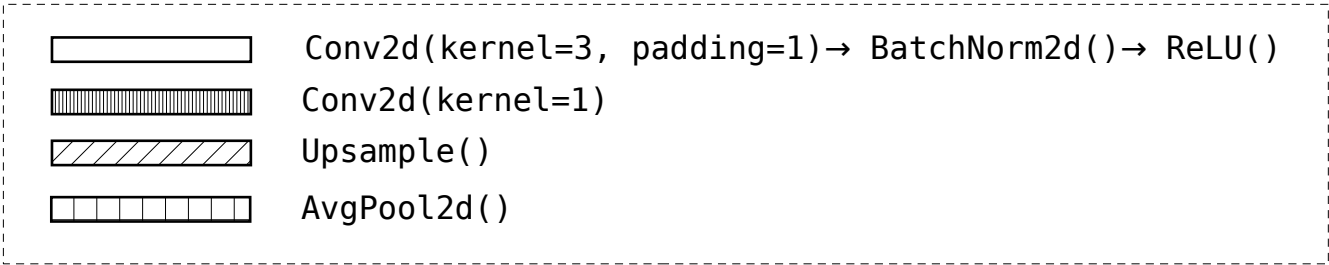


CNN Language Model



CNN Language Model



Results

- batch=50, learning_rate=0.02, training steps=4M, tokens=200M, TPP=24
- Scale up H, W, C
- vla.py --generate, compare to 1M, 8M weight transformer
- vla.py --vis

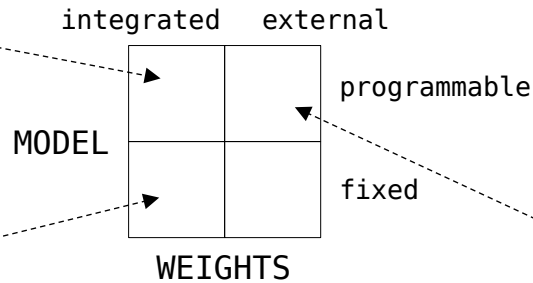
BACKUP

Inference Accelerator

Cerebras
Groq

Embedded Inference

integrated SRAM weights
fixed model
no software stack, weights only

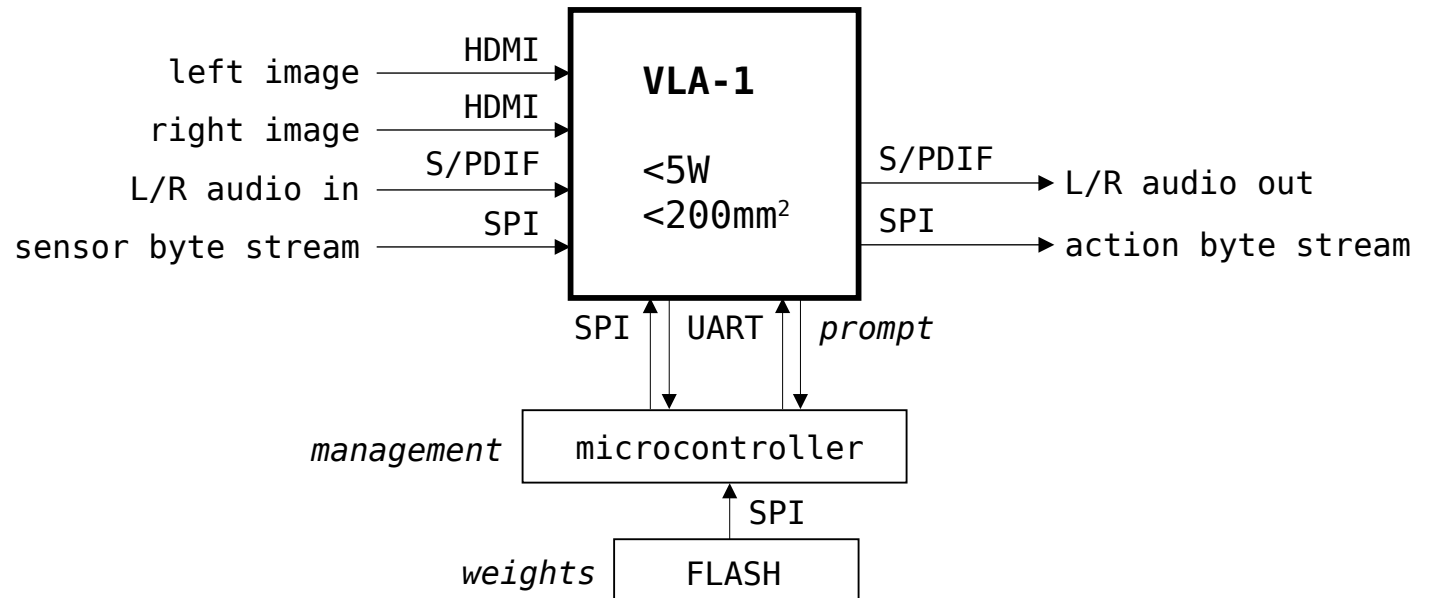


AI Accelerator (GPU)

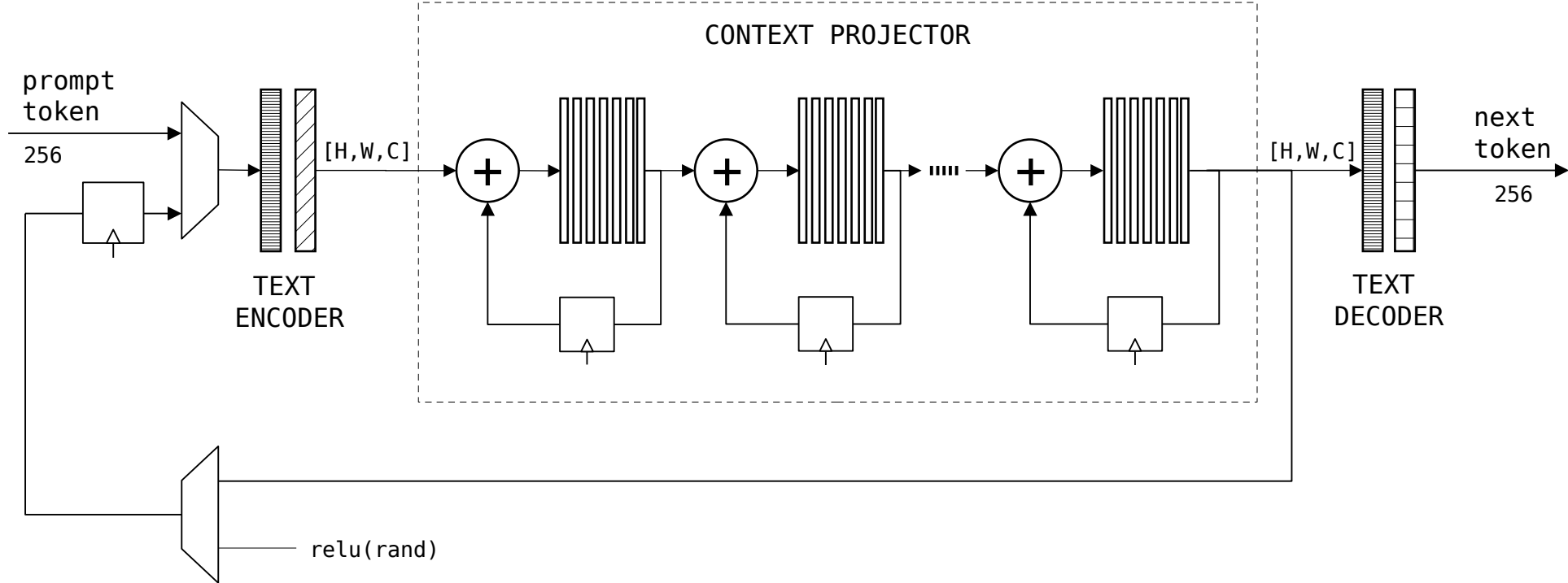
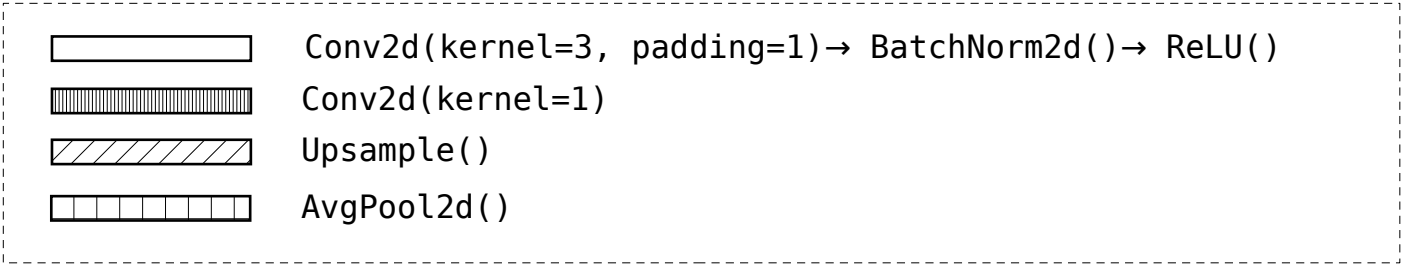
external DRAM weights
programmable model
Pytorch software stack

Embedded VLA

Autonomous robots
Interactive signage
Nuclear fusion reactors

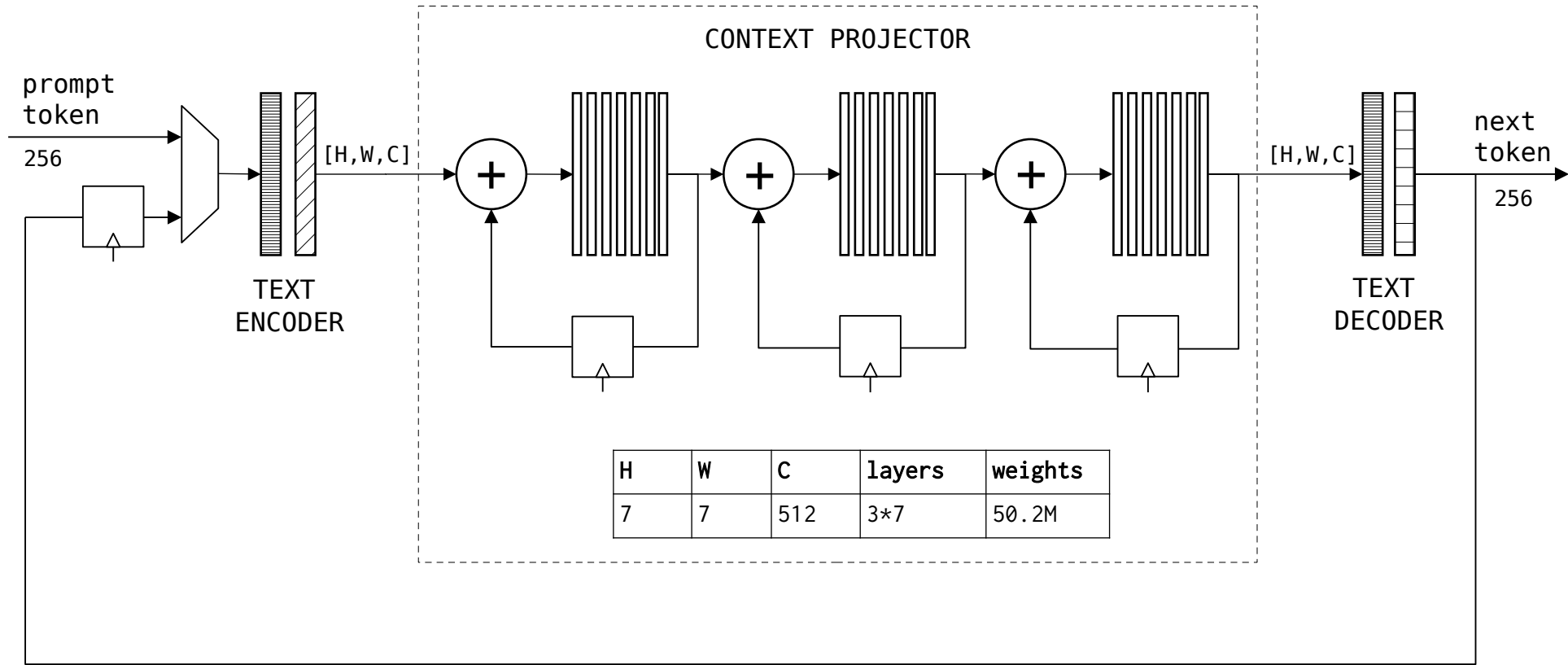
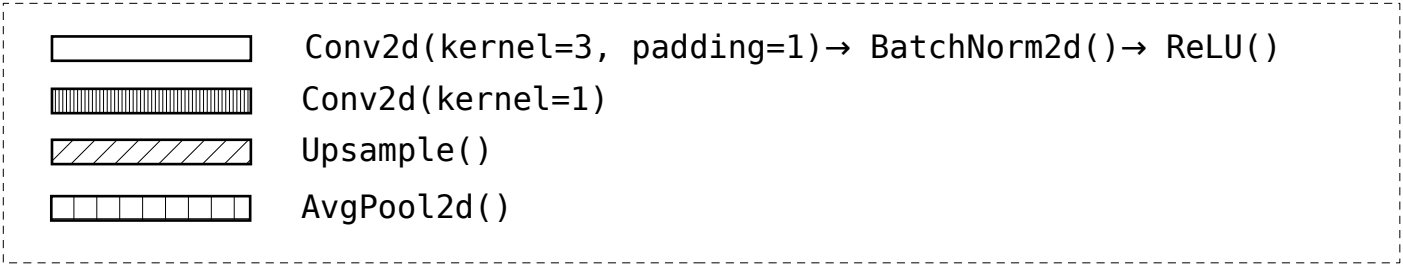


CNN Language Model

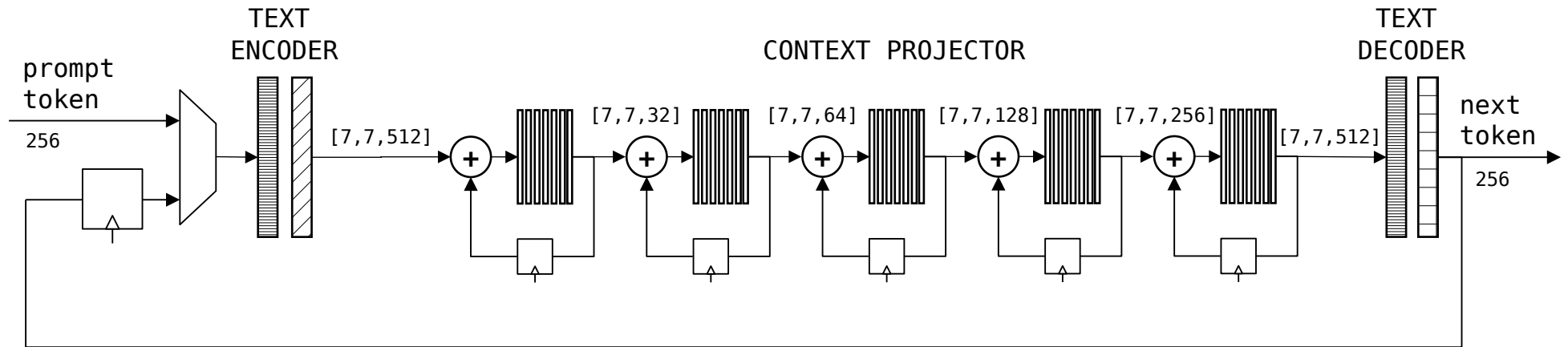
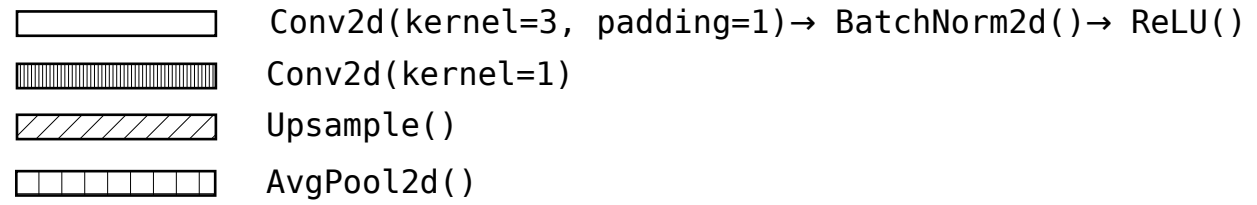


H	W	C	layers	weights
7	7	512	7*7	29.4M

CNN Language Model

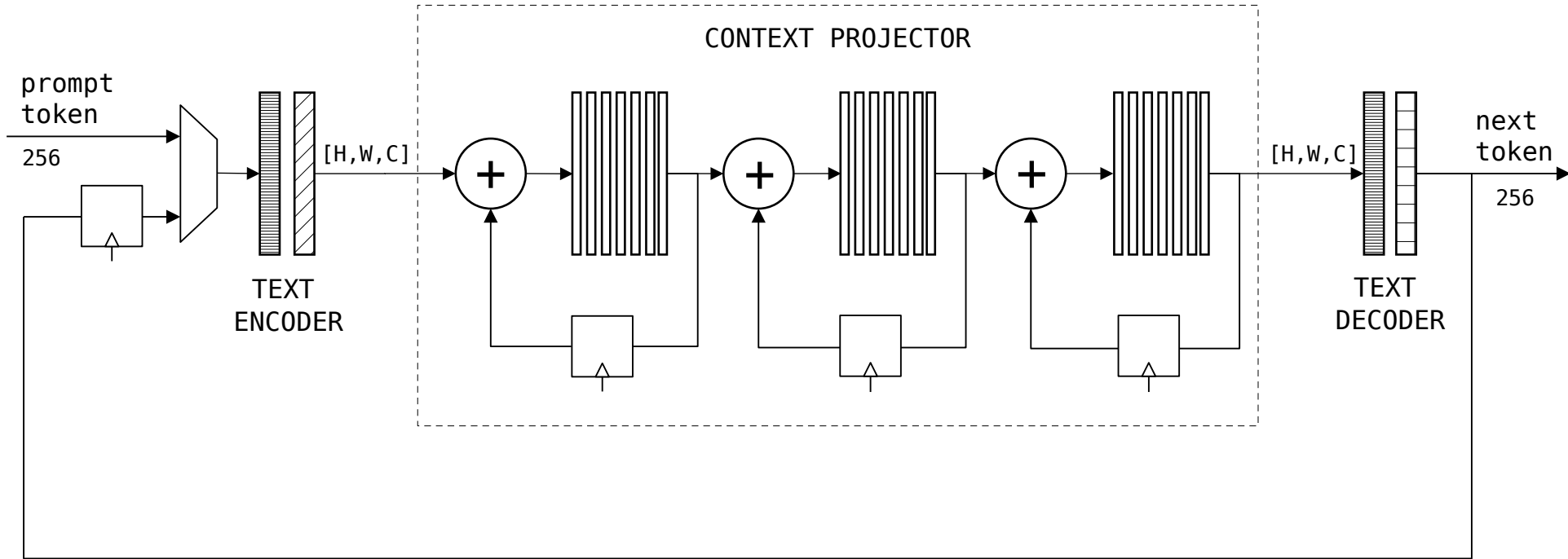
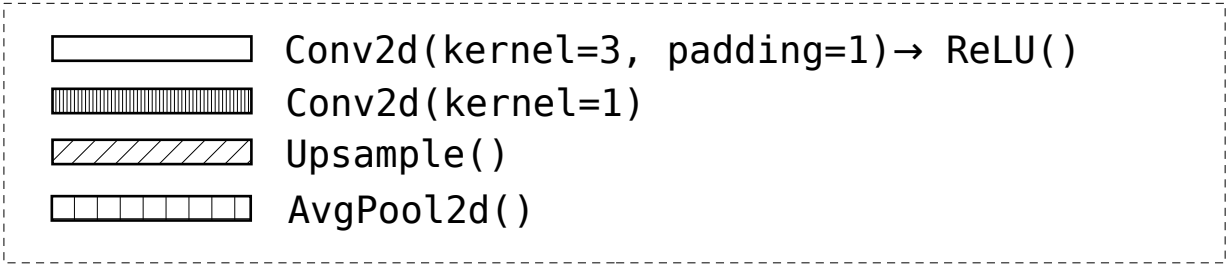


CNN Language Model



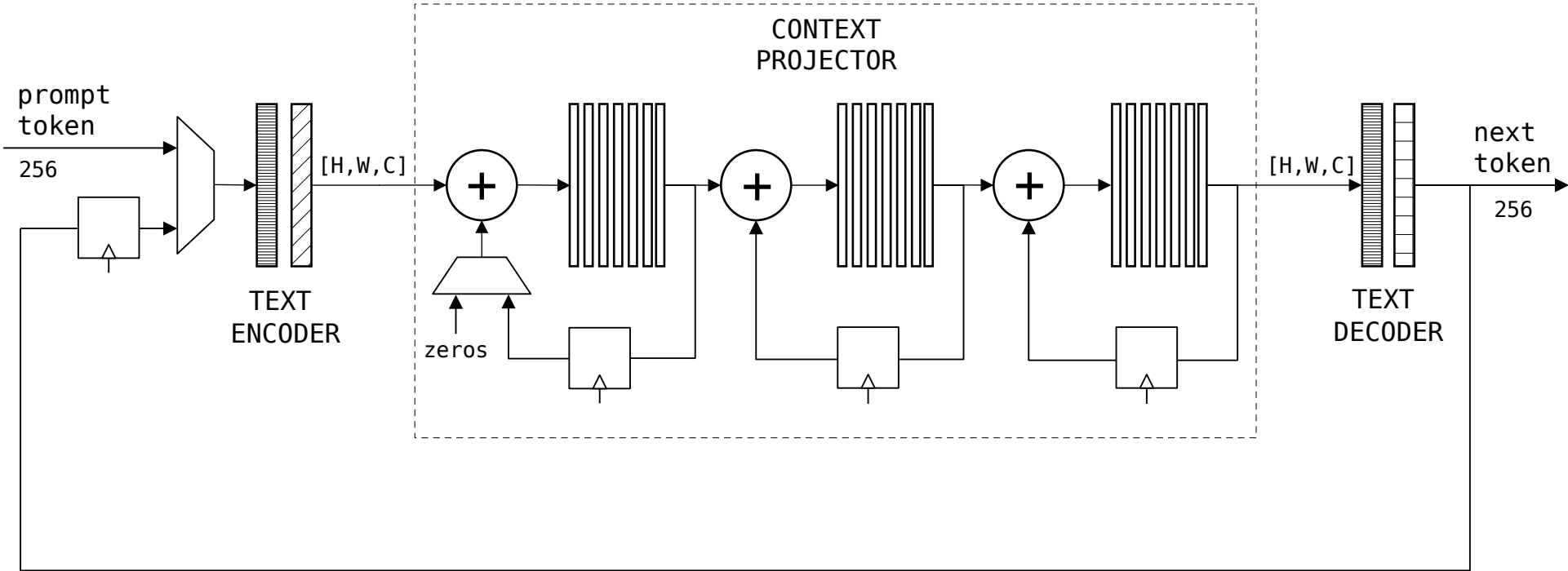
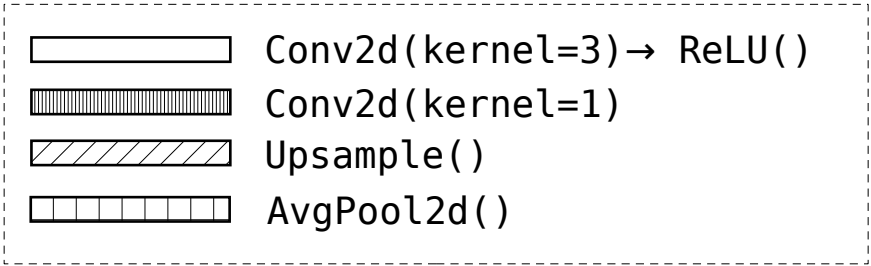
CNN Language Model

H	W	C	layers	weights
7	7	512	3*7	50.2M



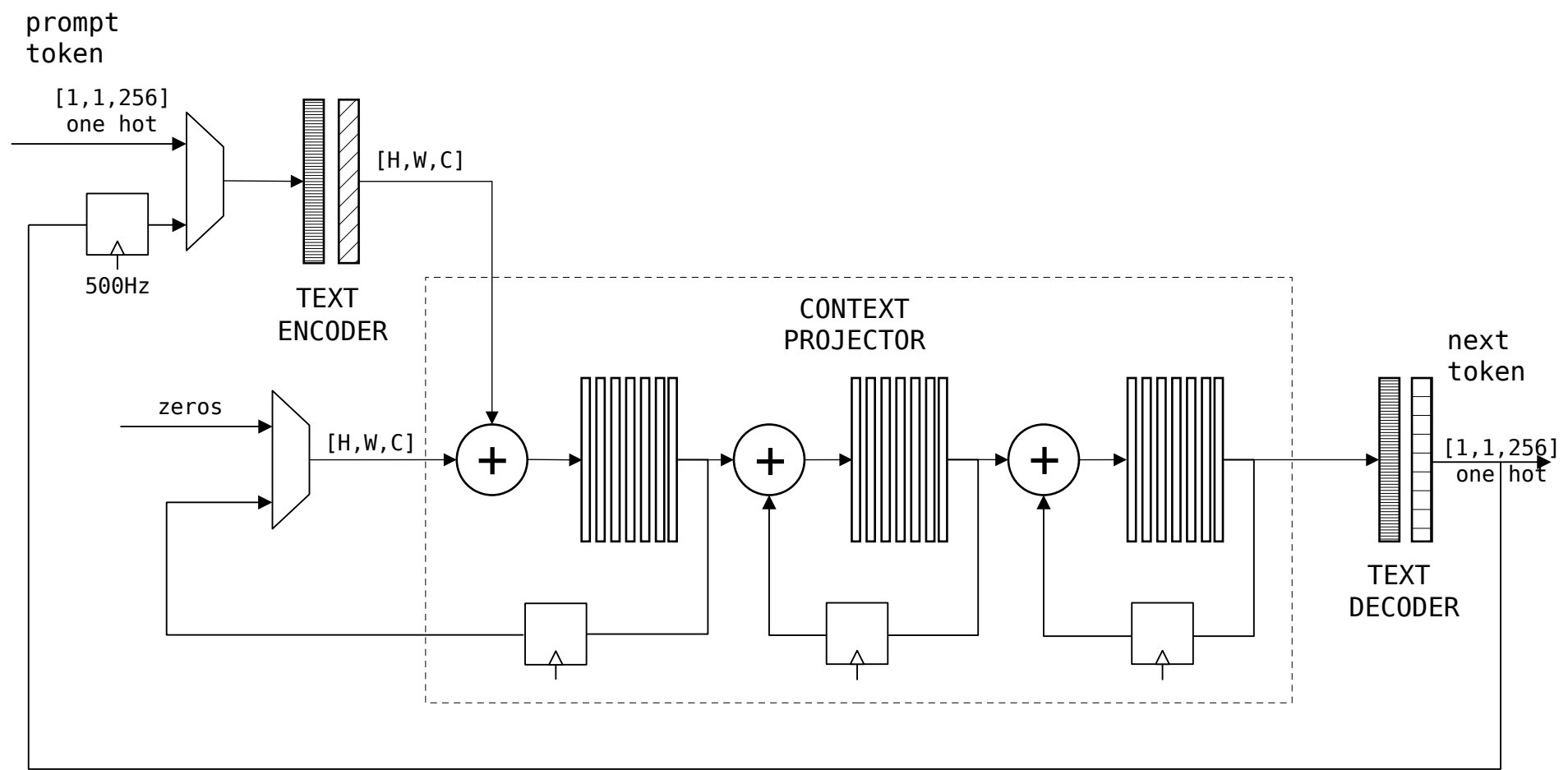
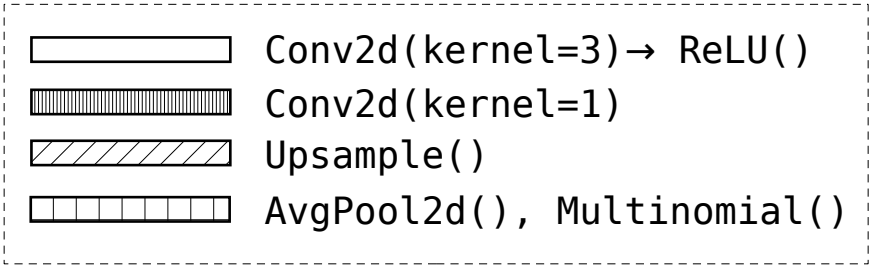
CNN Language Model

H	W	C	layers	weights
7	7	512	3*7	50.6M



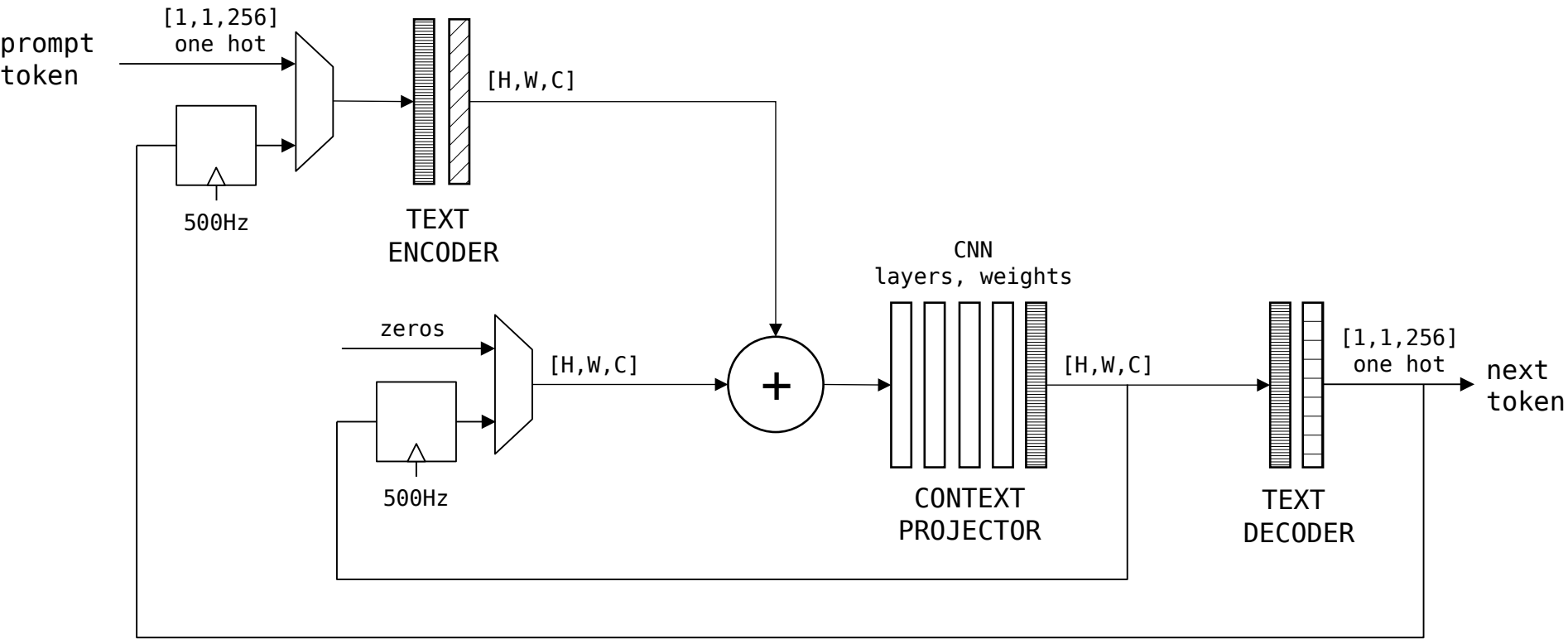
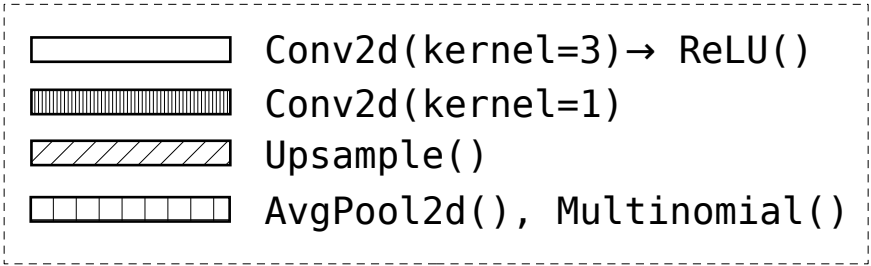
CNN Language Model

H	W	C	layers	weights
7	7	512	3*7	50.6M

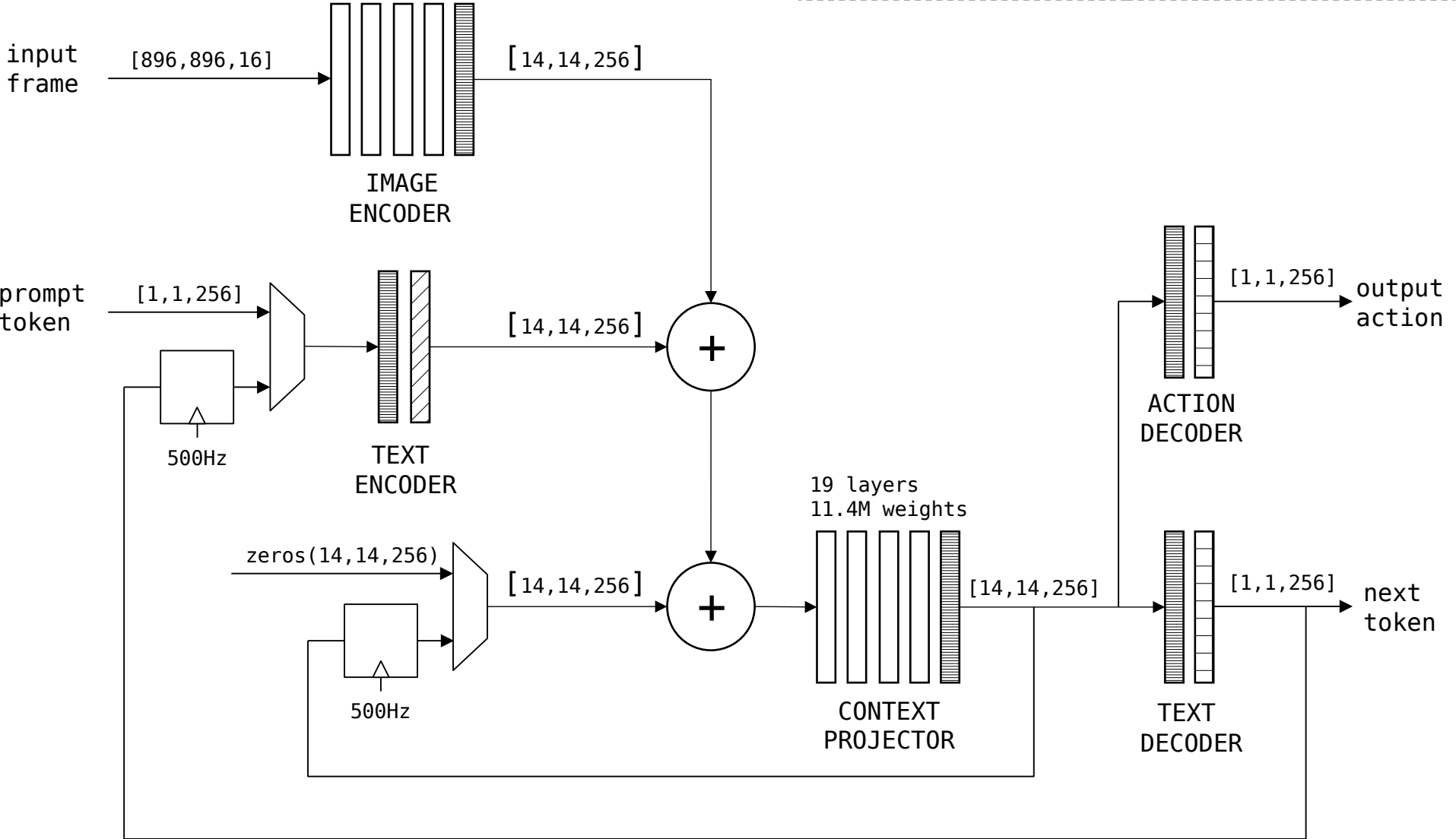
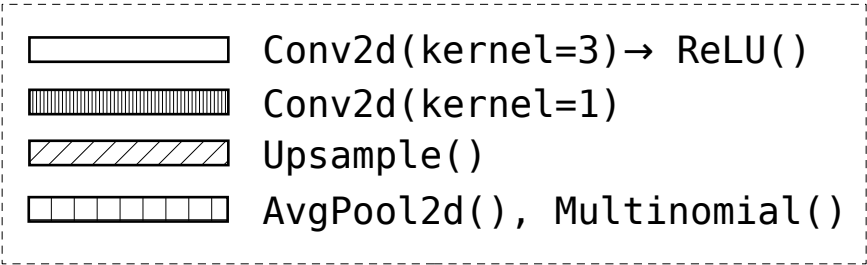


CNN Language Model

H	W	C	layers	weights
14	14	256	19	11.4M
56	56	16	224	528K



CNN Vision-Language-Action Model



integrated

Character level model at ~2B training characters

Words

Sentences

Basic grammar

Proper nouns

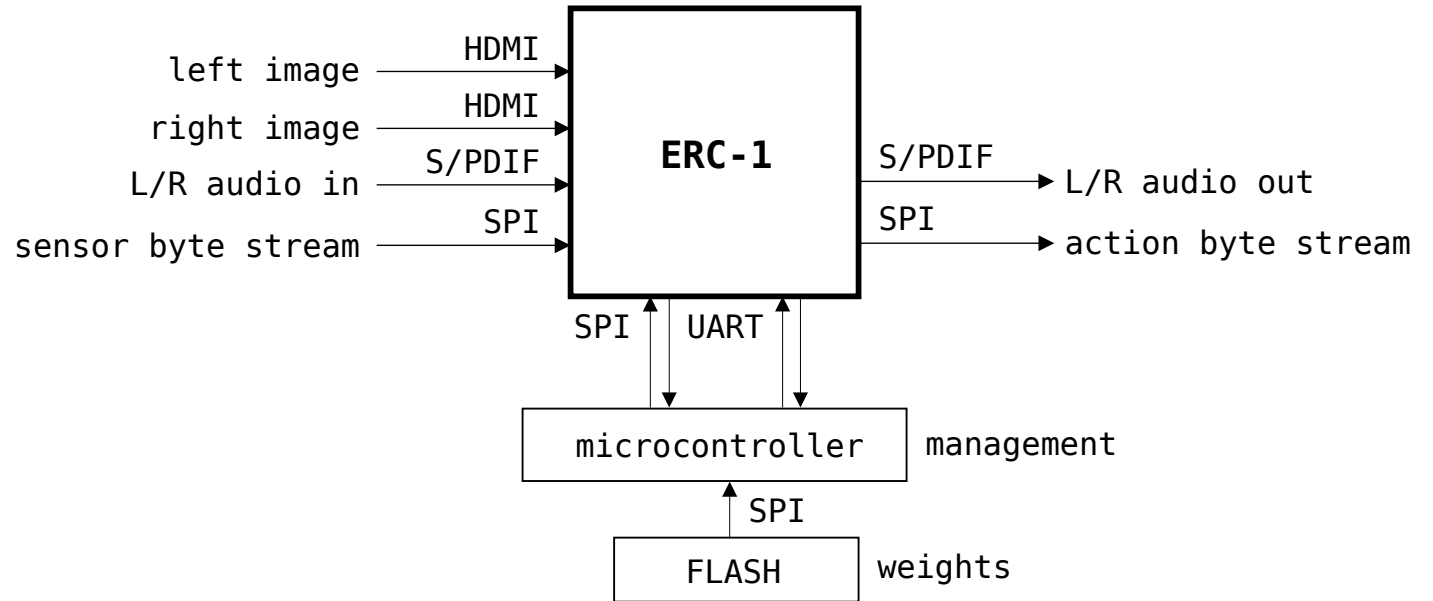
URL

Matched parenthesis

Numbers e.g. \$500 billion

Embedded Robot Controller

Autonomous robots
Interactive signage
Nuclear fusion reactors



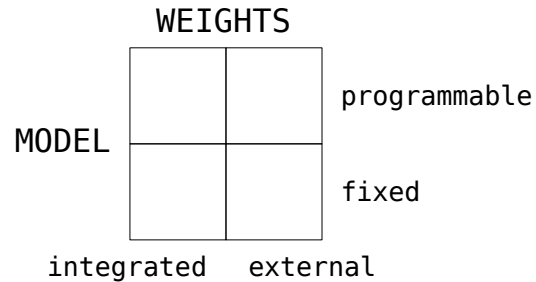
hardened components

integrated SRAM/ROM weights
fixed model

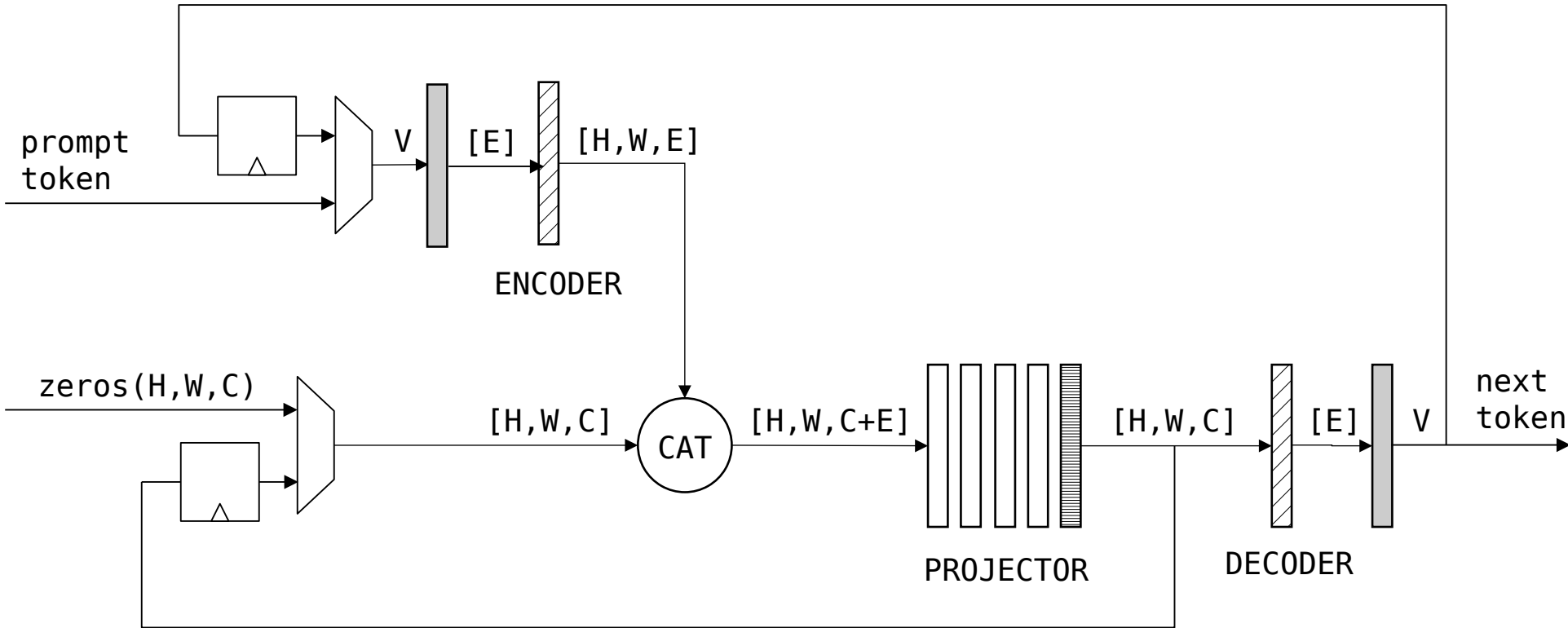


AI accelerator

external DRAM weights
programmable model



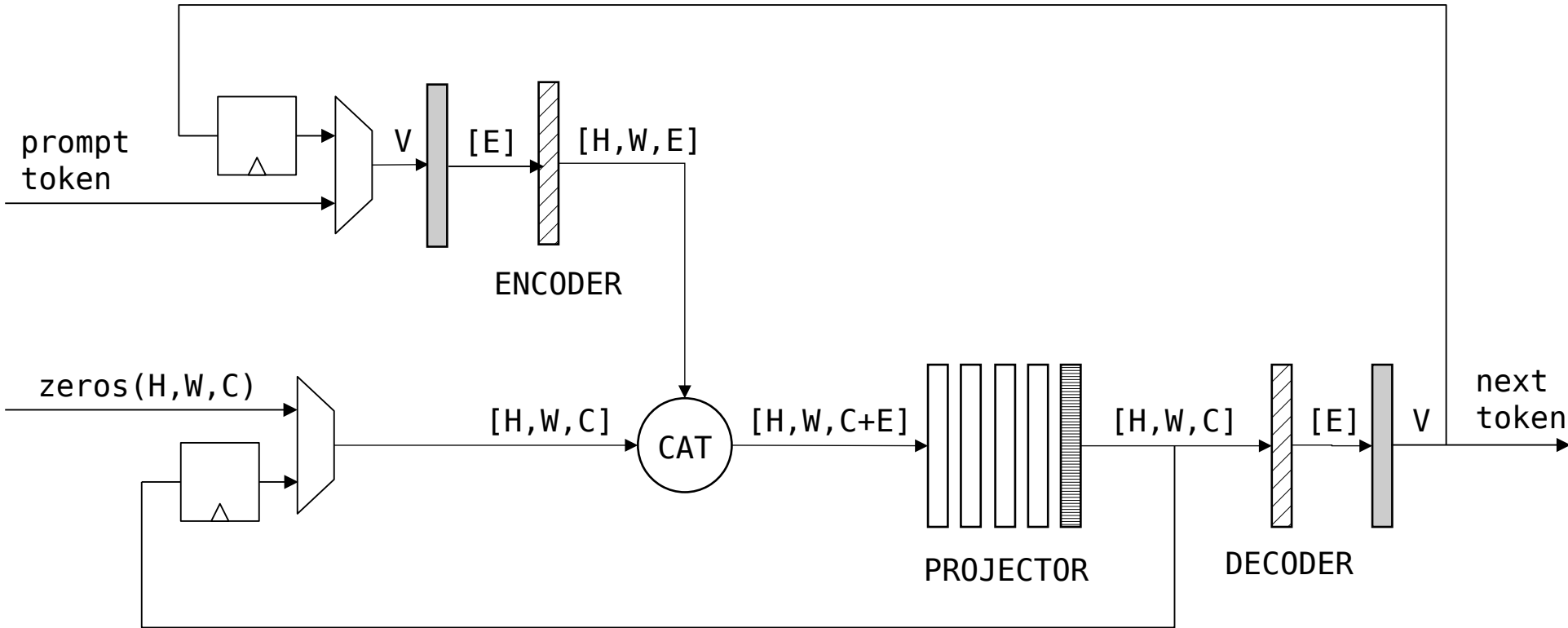
CNN Language Model

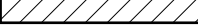


-  **Conv2d() → ReLU()**
-  **Conv2d()**
-  **Upsample(), AvgPool2d()**
-  **FROZEN**

V	50257
E	256
C	512
H,W	8,8

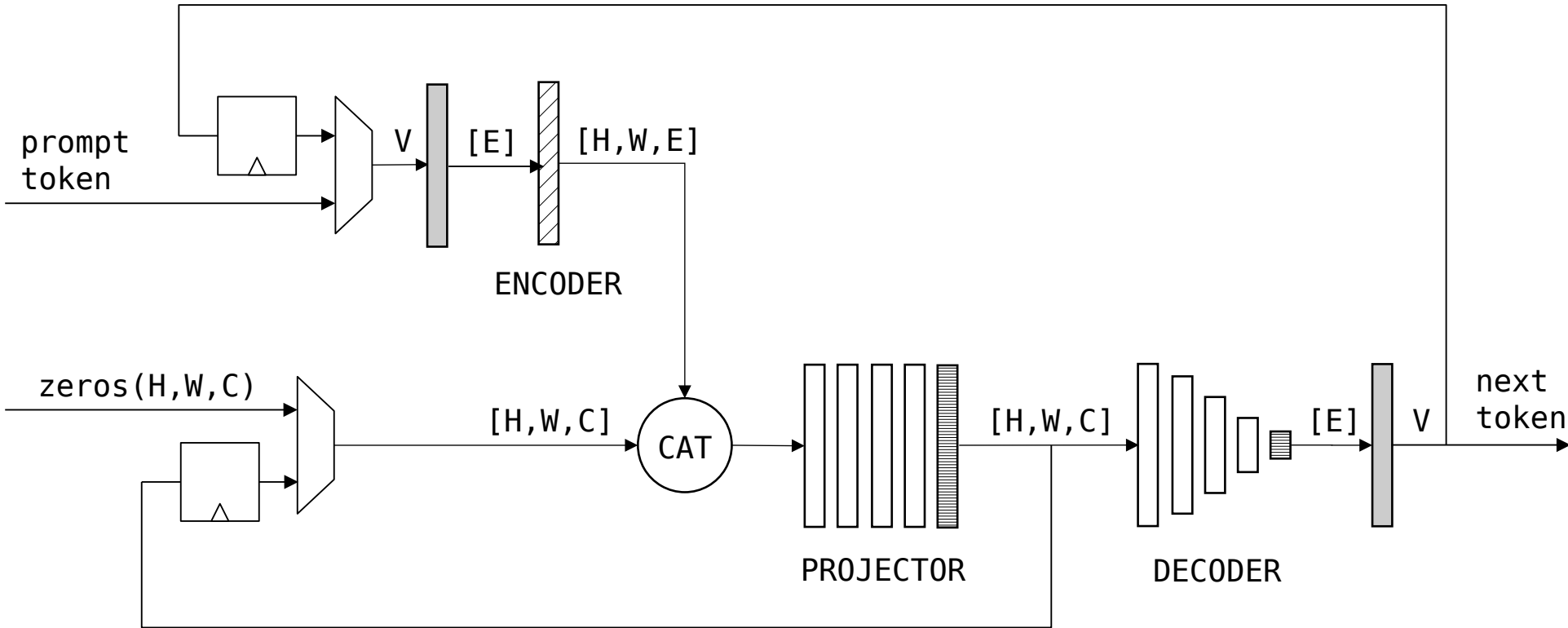
CNN Language Model

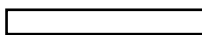


-  **Conv2d() → ReLU()**
-  **Conv2d()**
-  **Upsample(), AvgPool2d()**
-  **FROZEN**

V	50257
E	256
C	512
H,W	8,8

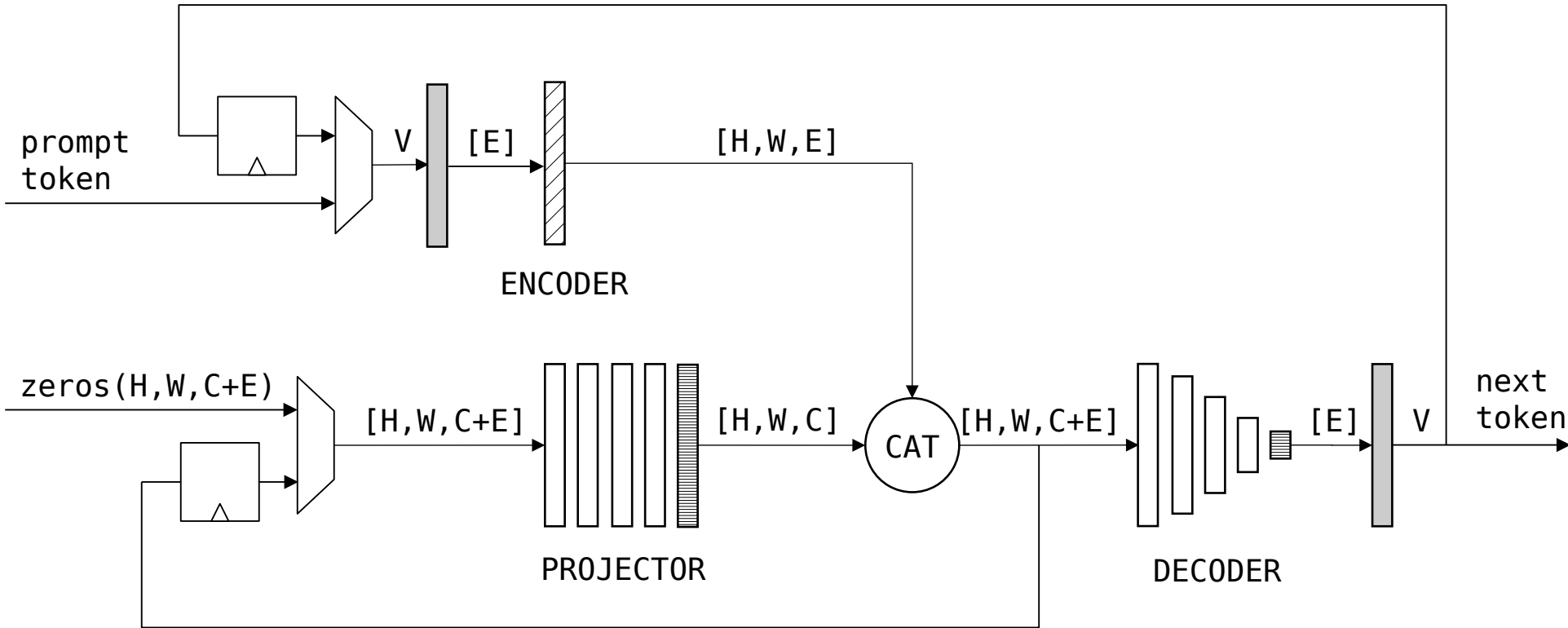
CNN Language Model

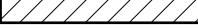


	Conv2d() → ReLU()
	Conv2d()
	Upsample()
	FROZEN

E	256
V	50257
C	256
H,W	14, 14

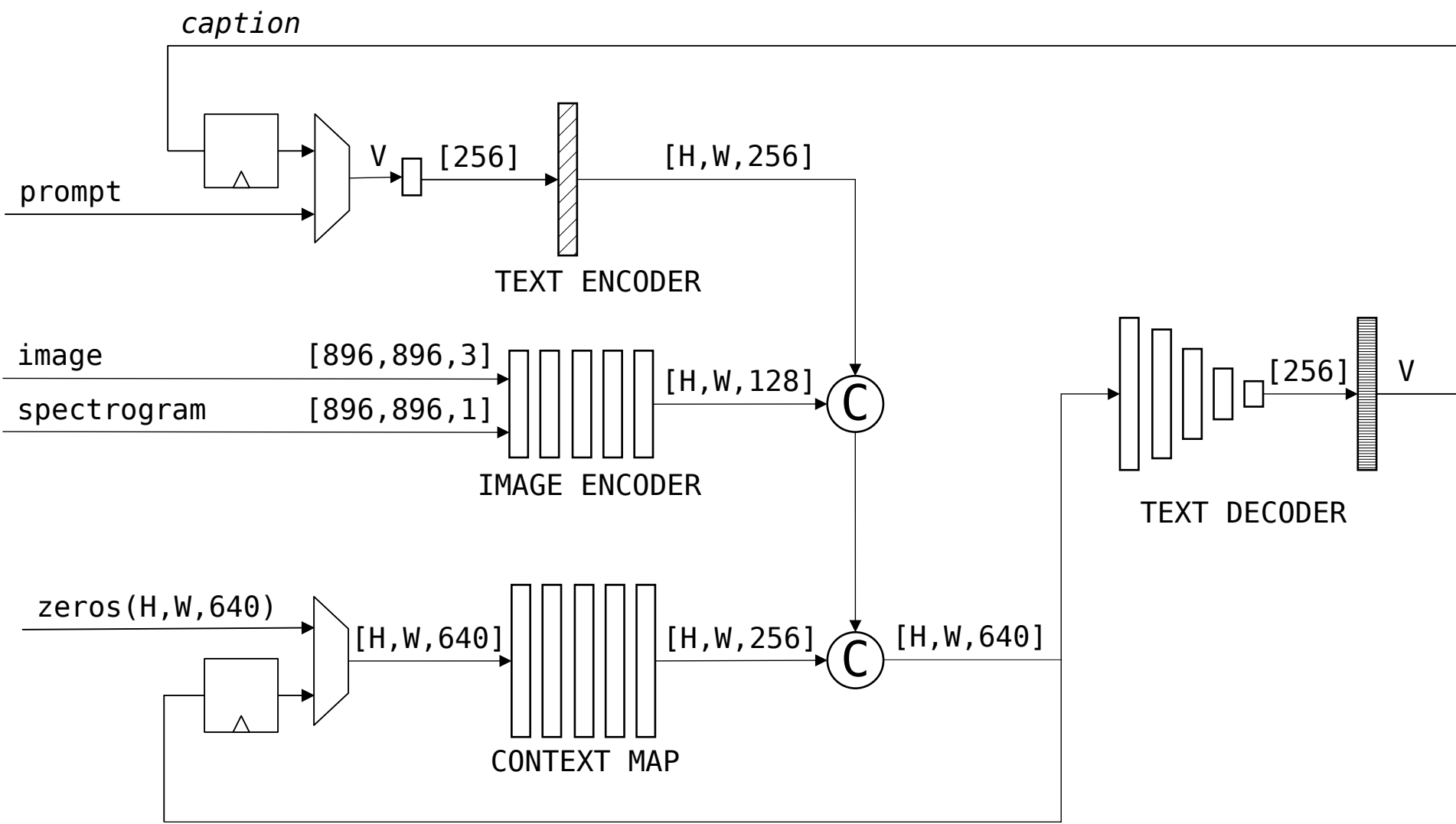
CNN Language Model



-  Conv2d() $\rightarrow$ ReLU()
-  Conv2d()
-  Upsample()
-  FROZEN

E	256
V	50257
C	256
H,W	14,14

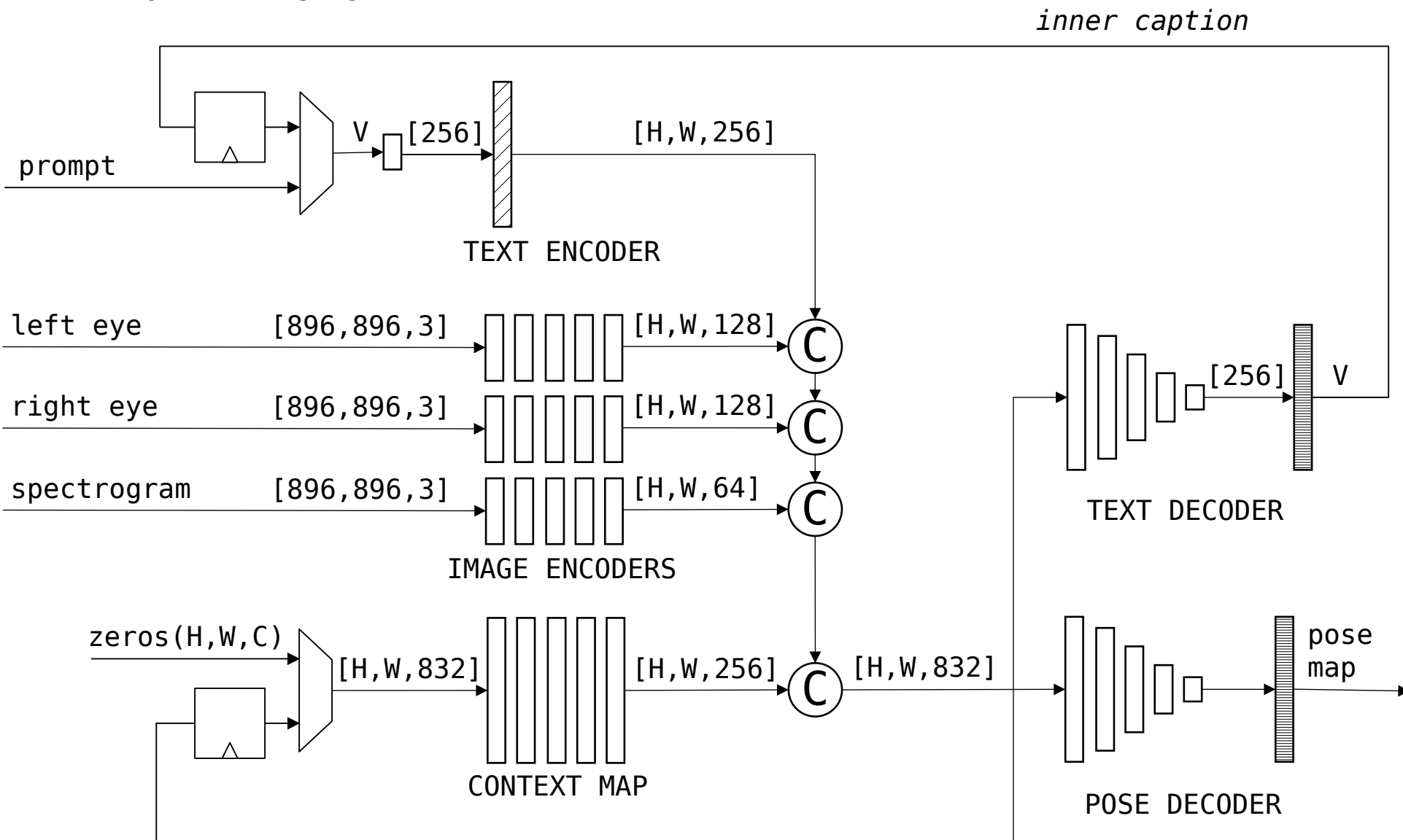
CNN Vision-Language Model

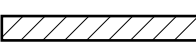


-  Conv2d() → ReLU()
-  Conv2d()
-  Upsample() → Conv2d()

H,W 32,32

CNN Perception-Language-Action Model



-  Conv2d() → BatchNorm2d() → ReLU()
-  Conv2d()
-  Upsample() → Conv2d()

H, W 32, 32

PHASE 1:

CNN Perception-Language-Action model (CNN-PLA), running on GPU
>20fps, <100ms latency, <400W

PHASE 2:

CNN-PLA model running on FPGA
>100fps, <15ms latency, <100W

PHASE 3:

CNN-PLA model running on ASIC at >300fps, <5ms latency, <10W

Seated humanoid :

1-DoF spine

2-DoF shoulder

2-DoF neck

6-DoF arms

6-DoF hands

Data collection :

VR teleoperated

spine/shoulder : swivel chair with up/down control

neck : VR gyroscope

arms/hands : TBD

ROBOT

896x896 RGB at 120Hz, left and right eye

896x896 spectrogram at 120Hz, audio and tactile signals

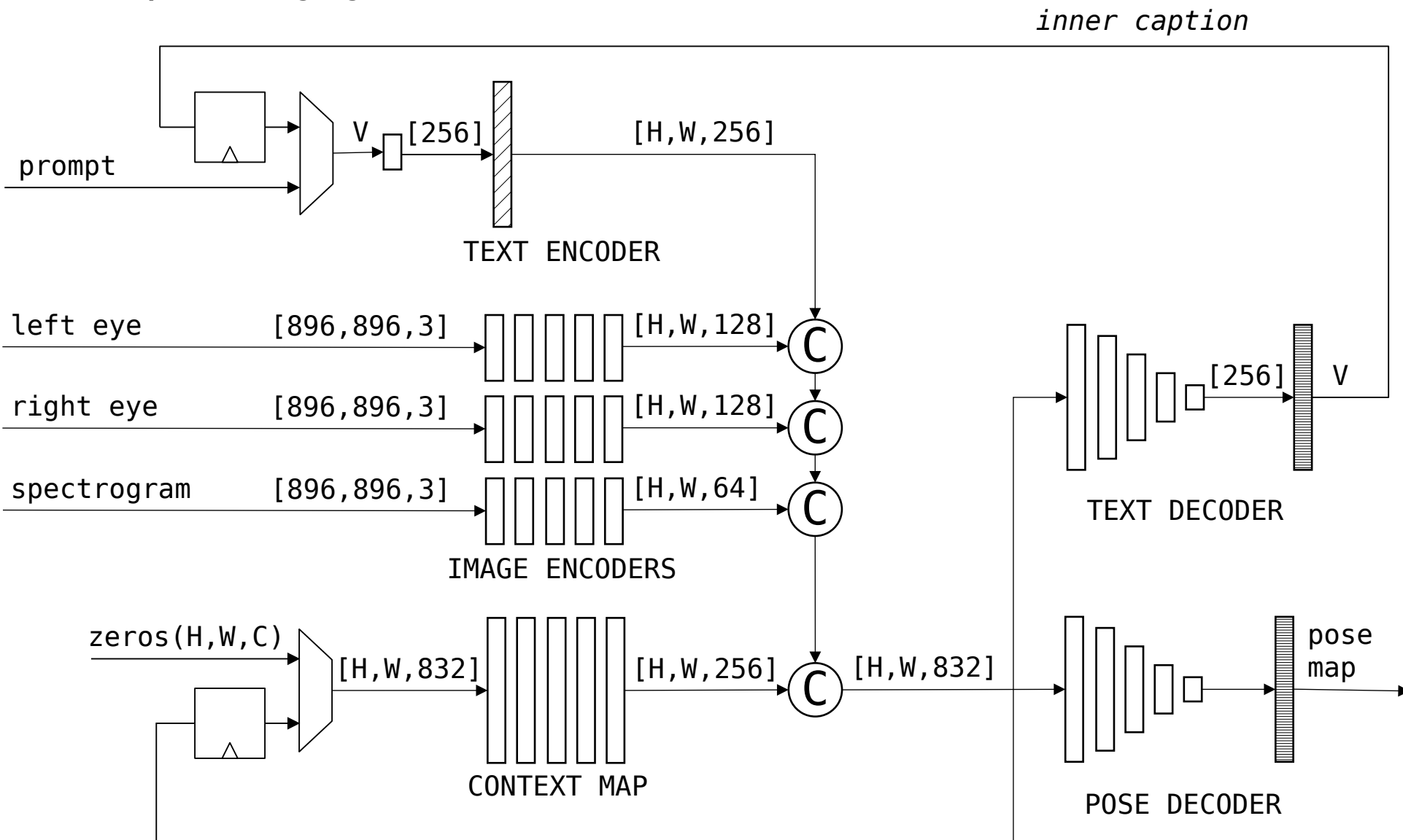
29-DoF target pose at 120Hz

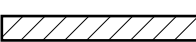
TELEOPERATOR

transcribed verbal commentary at 120 characters/s

synthetically captioned using open source VL model, 1 character/frame

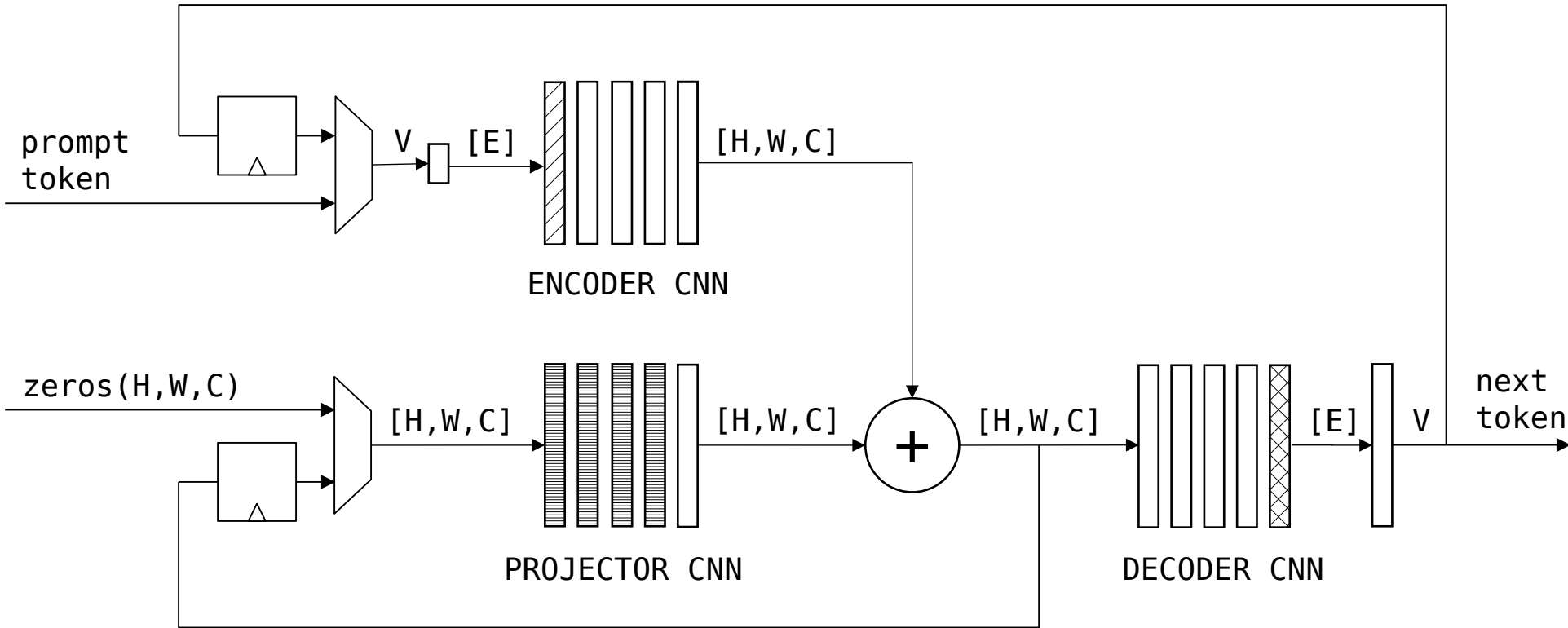
CNN Perception-Language-Action Model

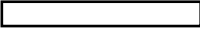


-  Conv2d() → BatchNorm2d() → ReLU()
-  Conv2d()
-  Upsample() → Conv2d()

H, W 32, 32

CNN Language Model



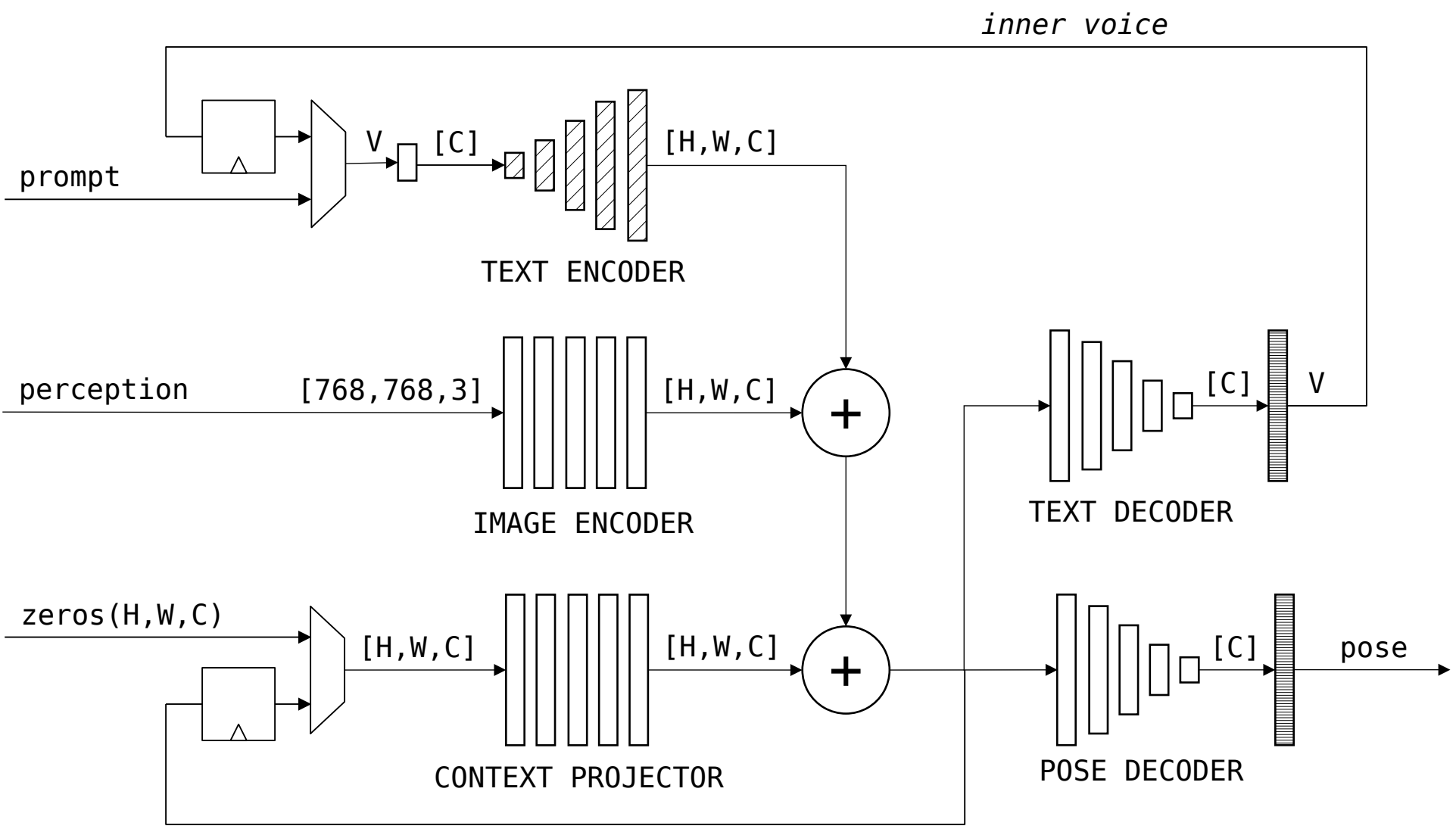
-  Conv2d() $\rightarrow$ ReLU()
-  Conv2d()
-  Upsample()
-  AvgPool()

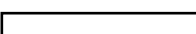
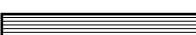
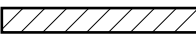
E	256
V	50257
C	384
H, W	14, 14

Devices

- TE256A : Token Encoder, vocab_size=256, ASCII
- TD256A : Token Decoder, vocab_size=256, ASCII
- CP384V : Context Projector, n_embd=384, VGG
- IE384R : Image Encoder, n_embd=384, Resnet
- TD
-

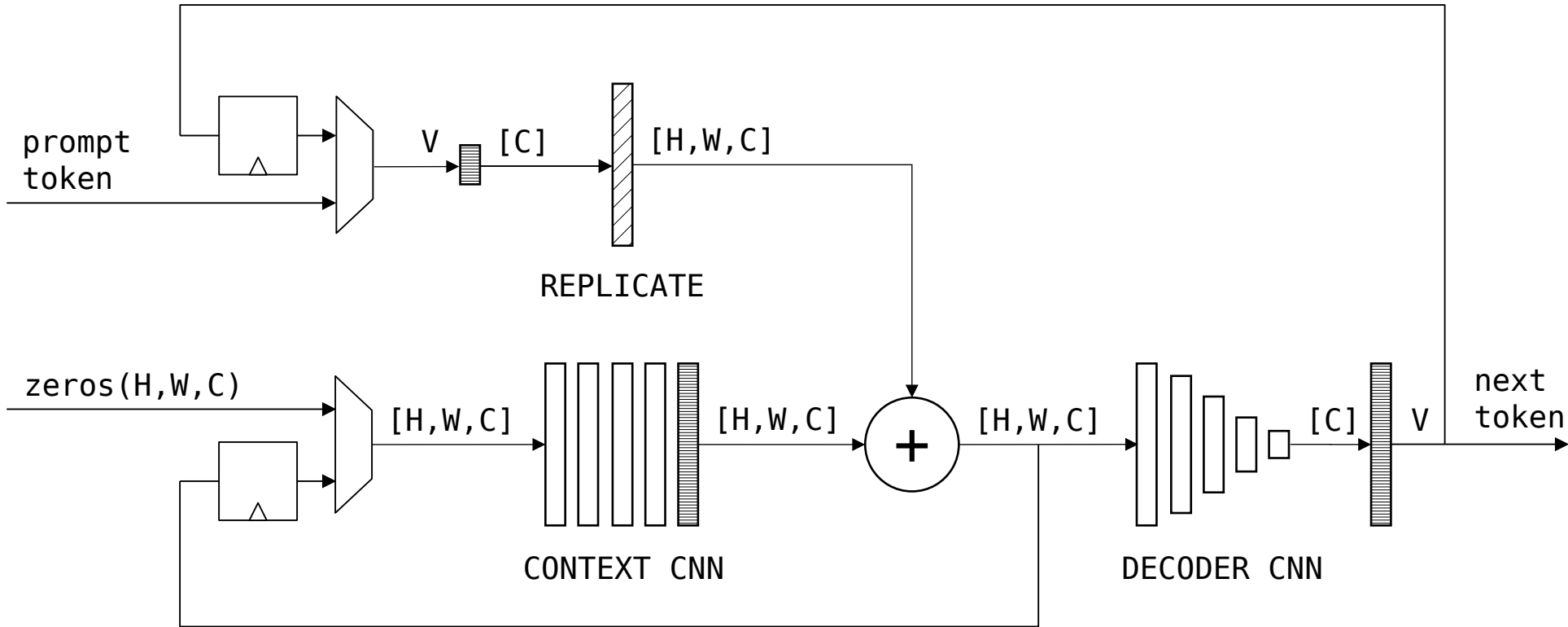
alt = lite-base-resnet-xeno

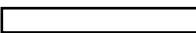
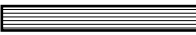
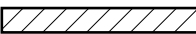


-  Conv2d(kernel=3)→ ReLU()
-  Conv2d(kernel=1)
-  Upsample(scale=3)→ Conv2d(kernel=3)→ ReLU()

V	256
C	384
H,W	81,81

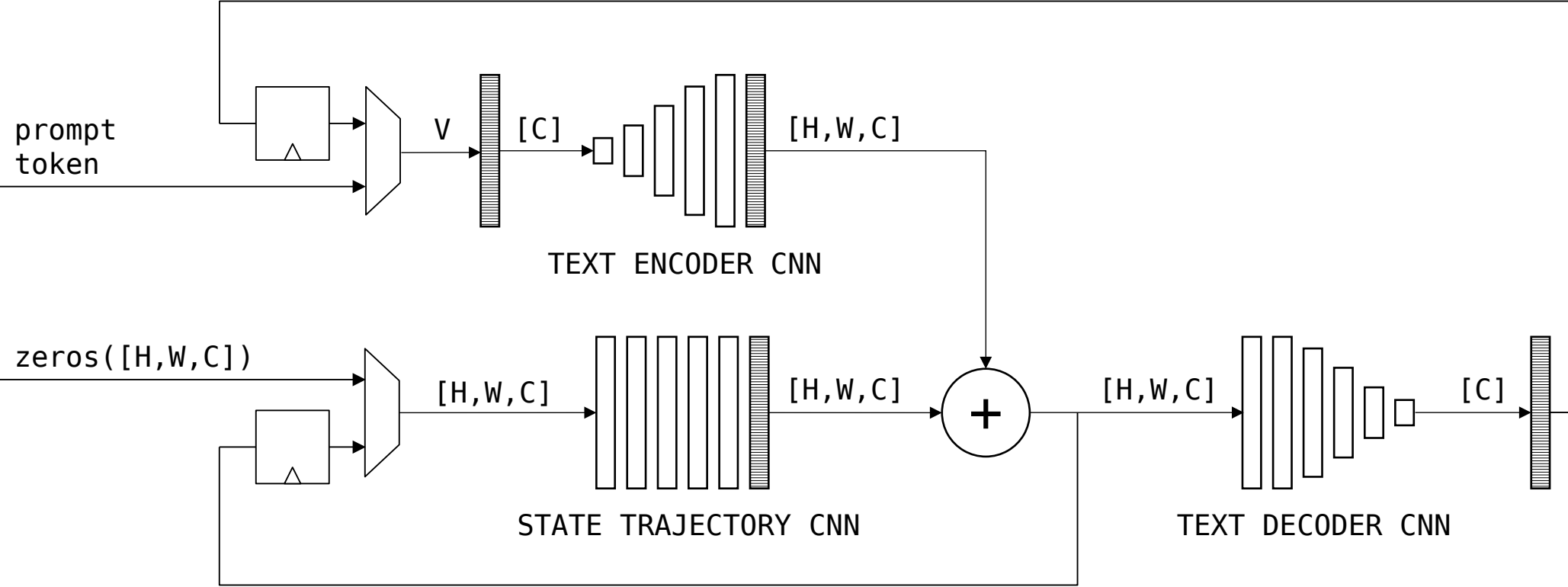
alt2



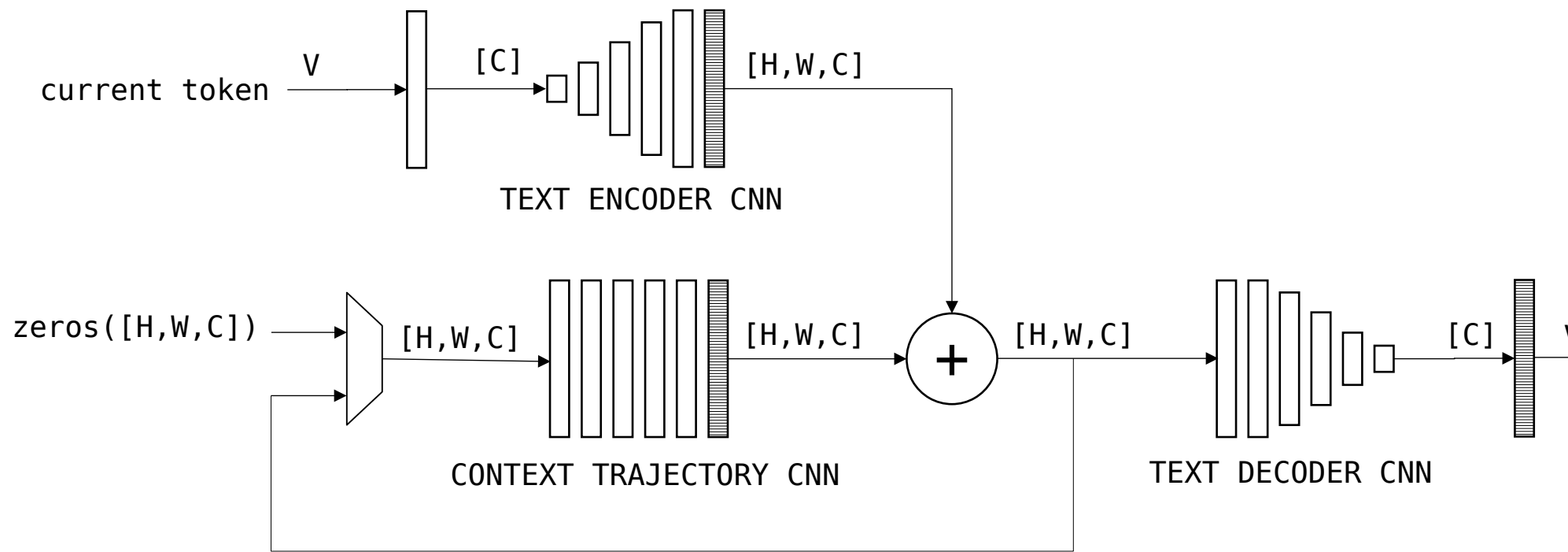
-  Conv2d(kernel=3) → BatchNorm2d() → ReLU()
-  Conv2d(kernel=1)
-  Upsample()

V	256
C	384
H,W	81,81

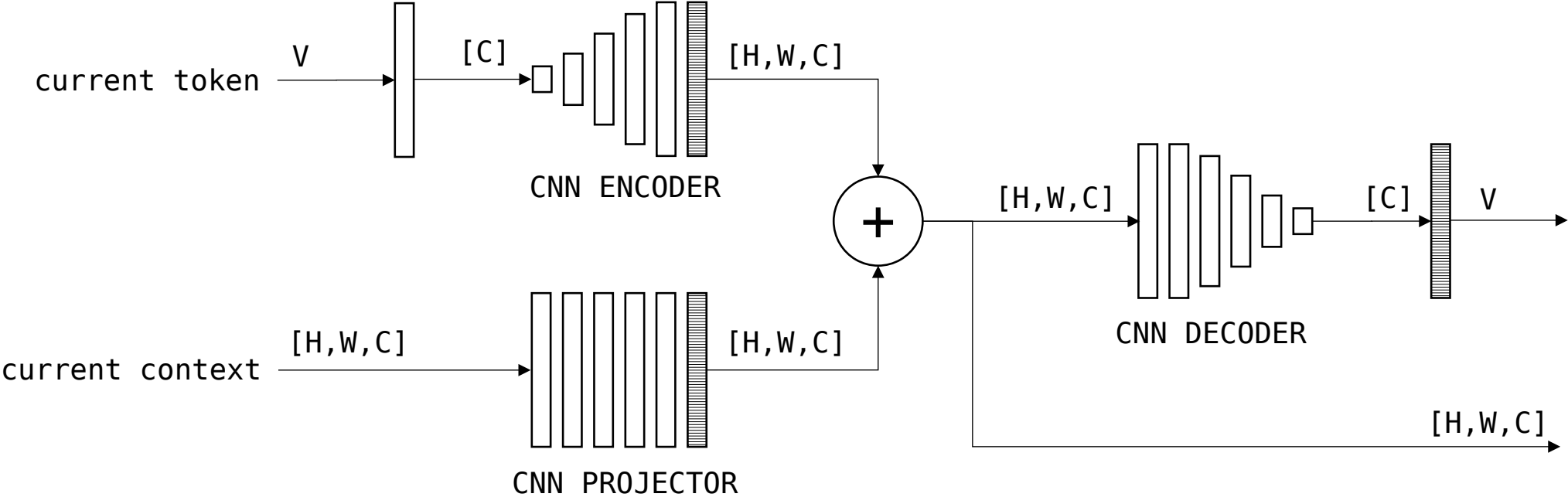
alt1



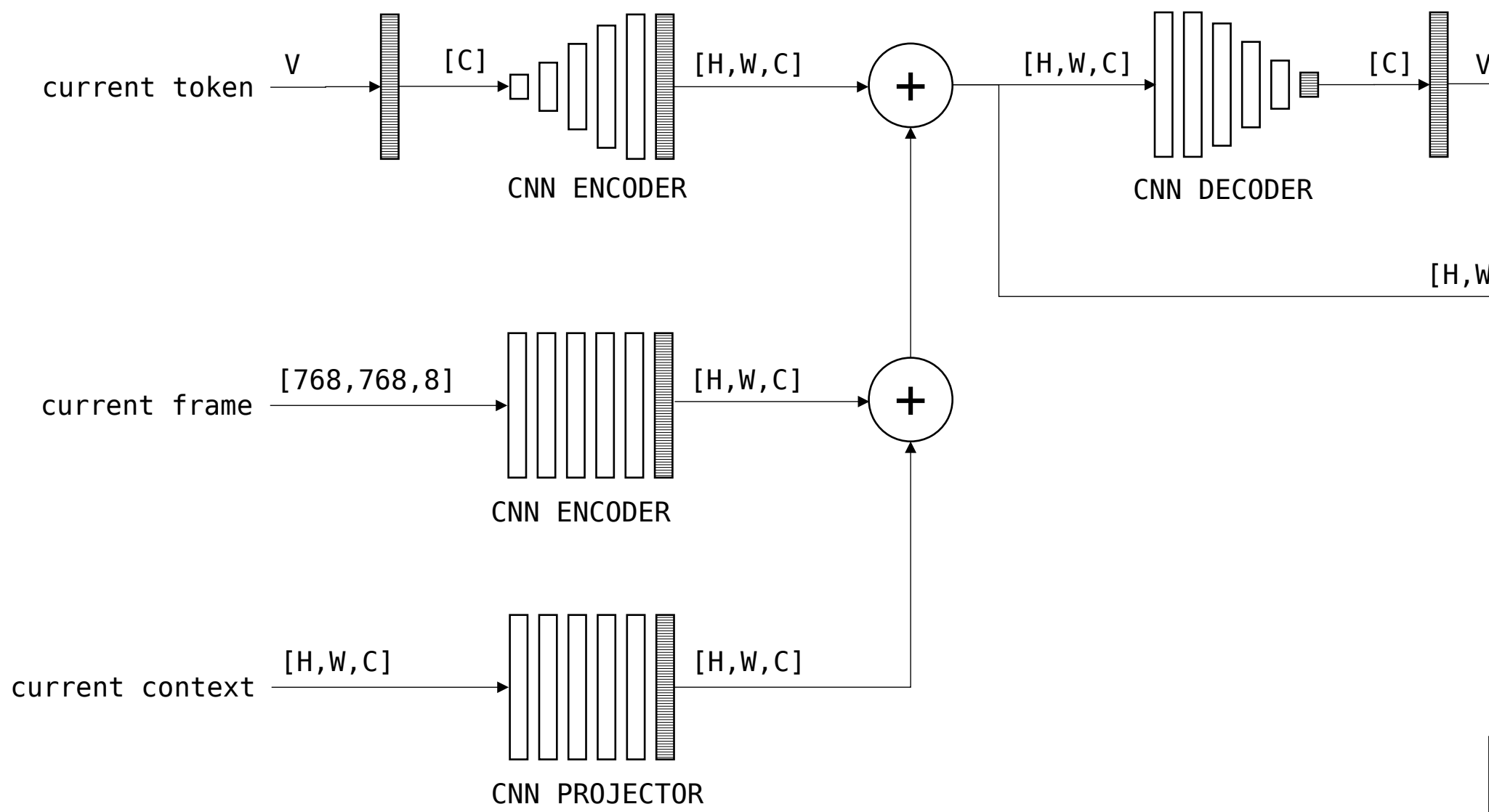
alt1



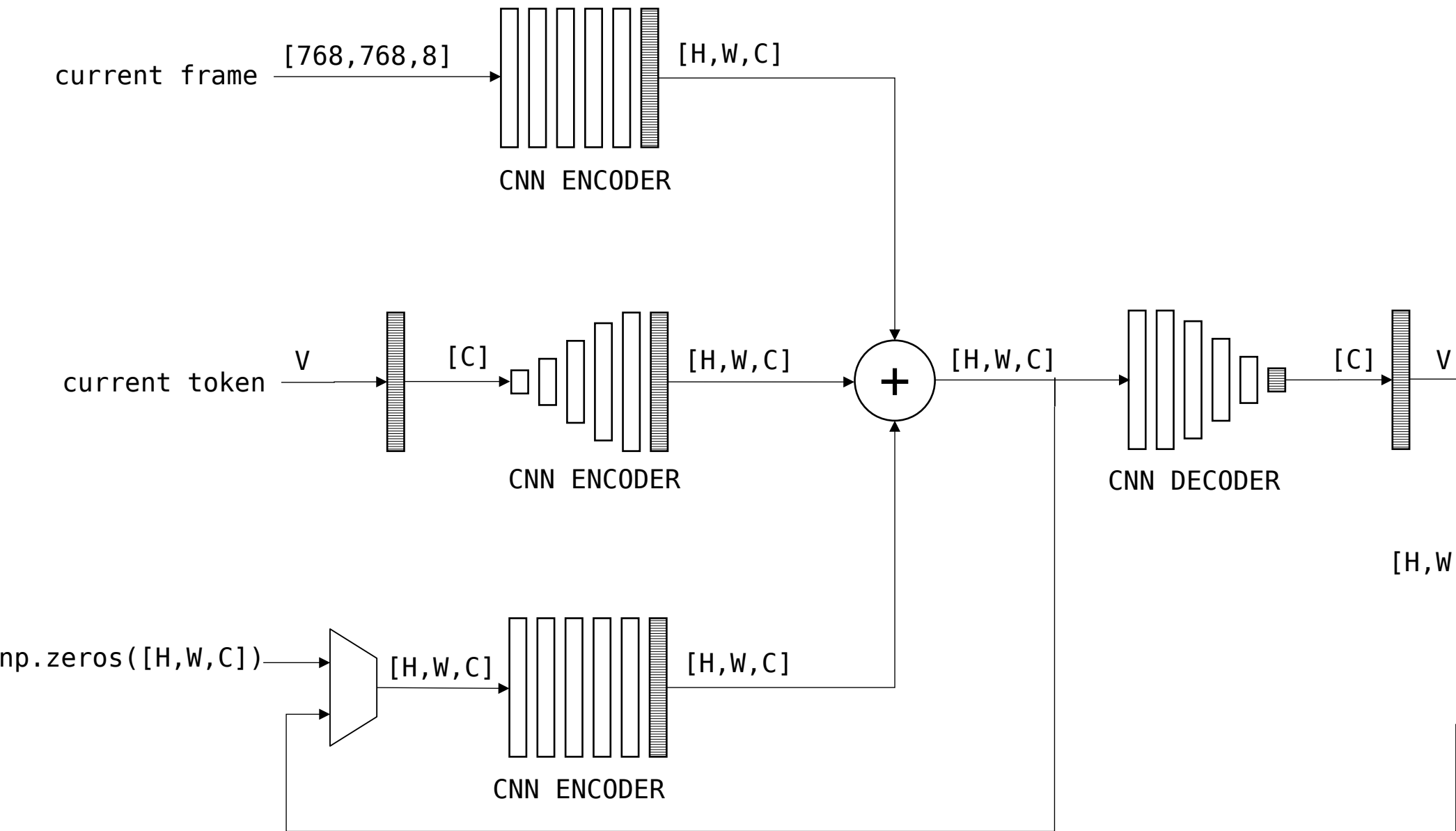
alt1



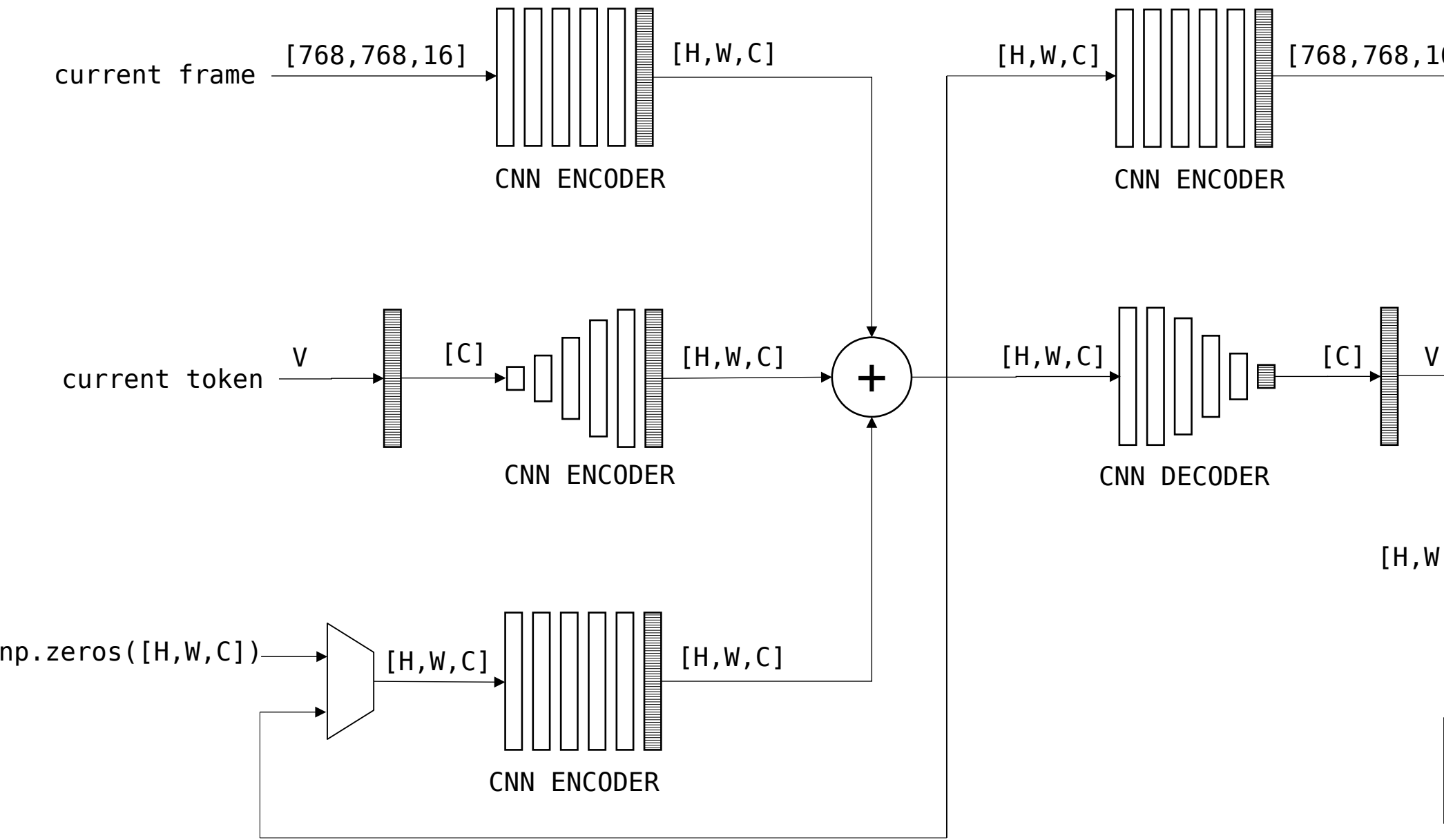
multimodal alt1



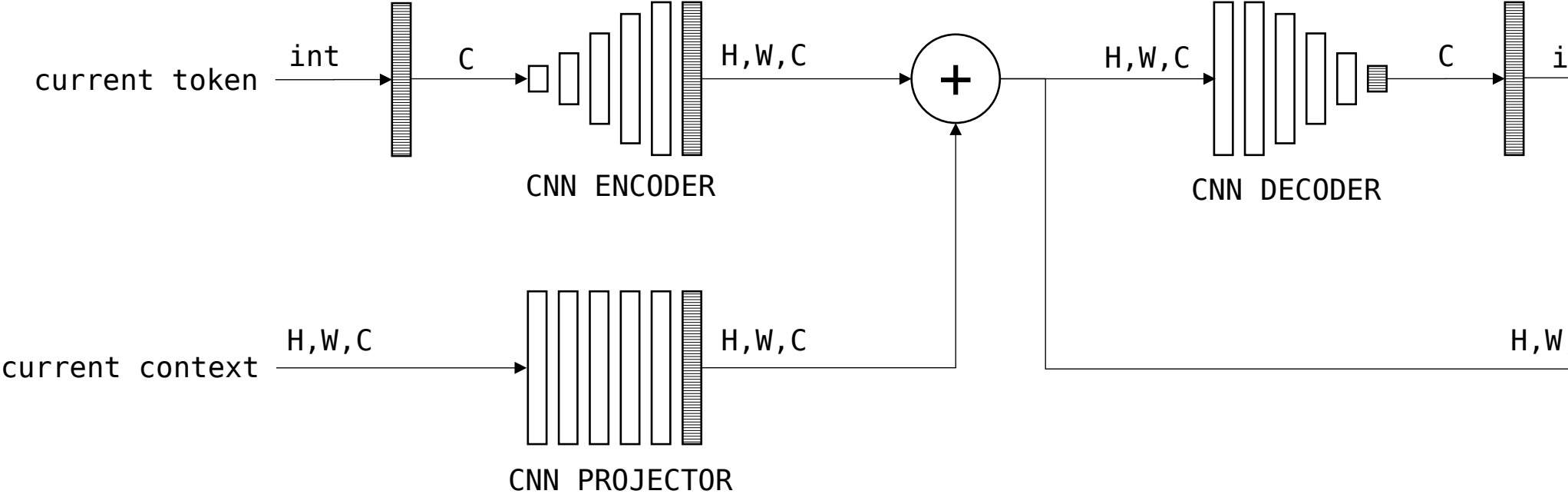
multimodal alt1

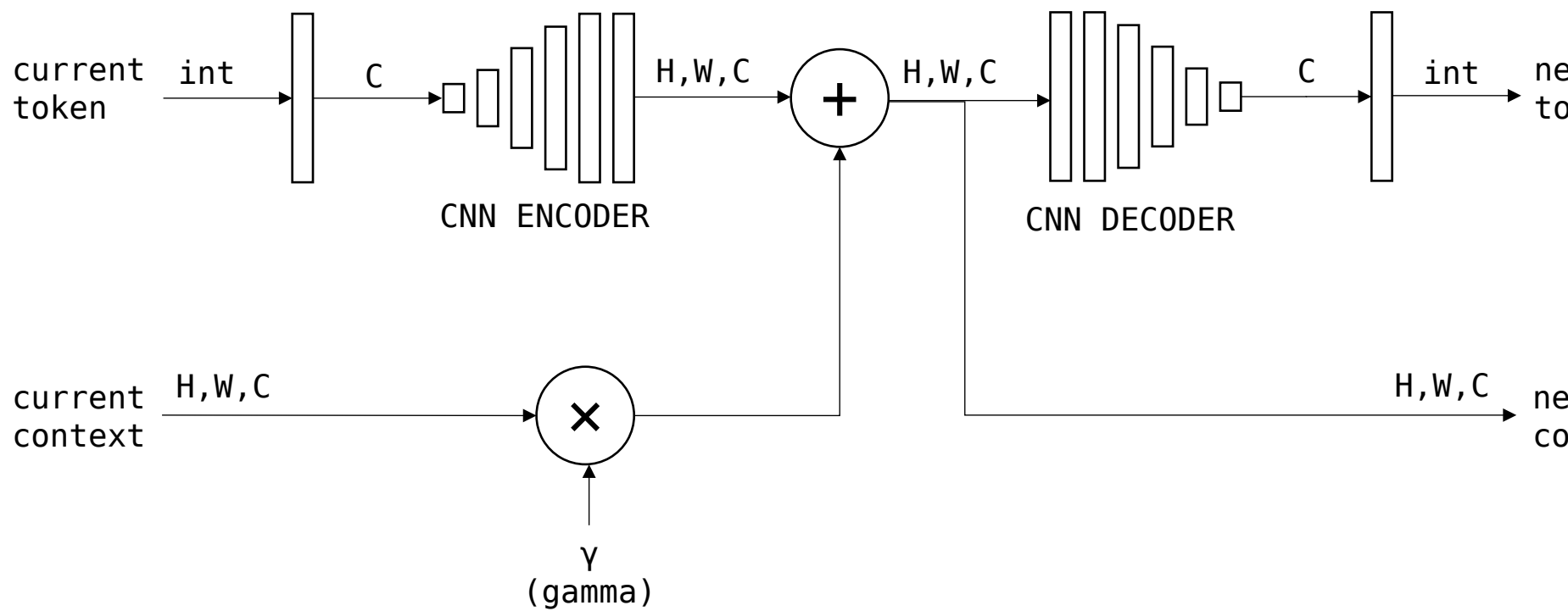


multimodal alt1

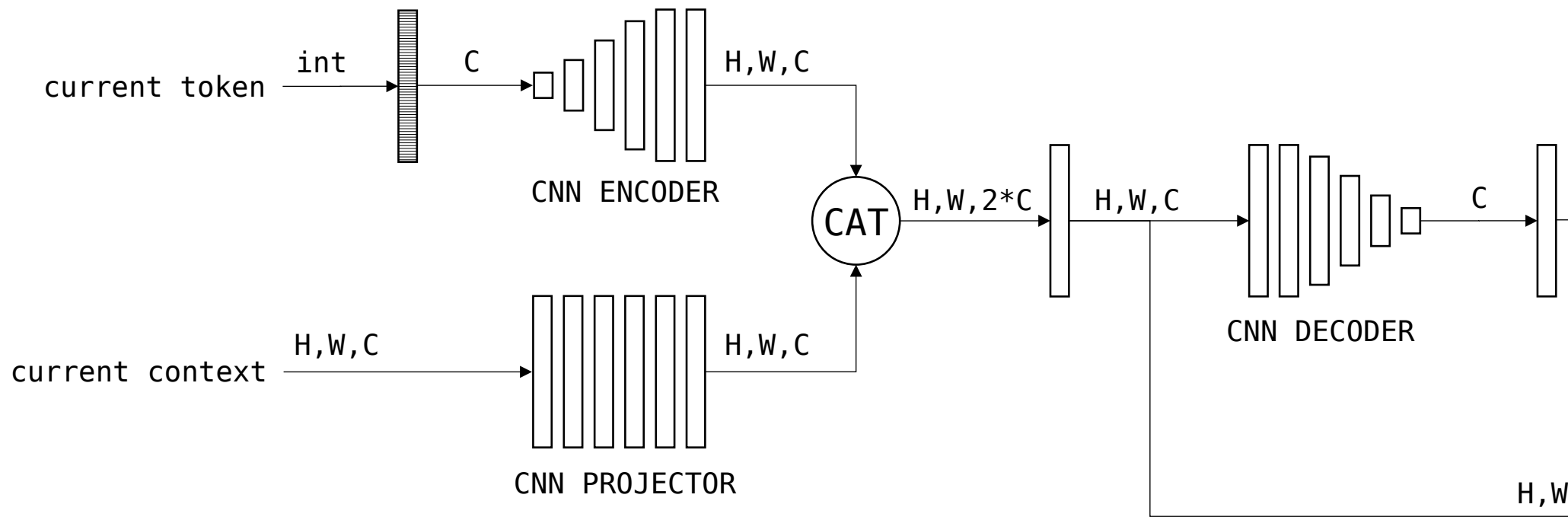


alt1





alt2



alt3

